



FACULTY OF
COMPUTING
& INFORMATICS

FINAL YEAR PROJECT INTERIM REPORT

771

Mobile App: Personalized Fitness Tracker & Smart
Workout Planner

1211111950

ADAM FIKRI BIN MOHD LOTFI

BACHELOR IN COMPUTER SCIENCE (HONS.) SOFTWARE
ENGINEERING

FEBRUARY 2026

771
Mobile App: Personalized Fitness Tracker & Smart
Workout Planner

BY
1211111950
ADAM FIKRI BIN MOHD LOTFI

PROJECT INTERIM REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENT FOR THE DEGREE OF
BACHELOR IN COMPUTER SCIENCE (HONS.) SOFTWARE ENGINEERING

in the
Faculty of Computing and Informatics

MULTIMEDIA UNIVERSITY
MALAYSIA

FEBRUARY 2026

Copyright

© <Year of Report submission> Universiti Telekom Sdn. Bhd. ALL RIGHTS RESERVED.

Copyright of this report belongs to Universiti Telekom Sdn. Bhd. as qualified by Regulation 7.2 (c) of the Multimedia University Intellectual Property and Commercialisation Policy. No part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Universiti Telekom Sdn. Bhd. Due acknowledgement shall always be made of the use of any material contained in, or derived from, this report.

Declaration

I hereby declare that the work has been done by myself and no portion of the work contained in this report has been submitted in support of any application for any other degree or qualification of this or any other university or institution of learning.

Name of candidate: ADAM FIKRI BIN MOHD LOTFI

Faculty of Computing & Informatics

Multimedia University

Date: 11: 02: 2026

Acknowledgements

To begin, I would like to express my utmost thanks to Allah S.W.T. for giving me the strength, patience, and guidance that I need to successfully complete my Final Year Project 1 (FYP1).

In particular I would like to thank my supervisor Madam Nur Haifa binti Mohd Fathil for her continuous support, guidance and constructive feedback during the course of this project. Your encouragement and expertise played a huge part in shaping the direction of this research.

I would also like to thank the Faculty of Computing and Informatics at Multimedia University for providing me with the resources and learning environment required to carry out this project.

I would like to thank each and every one of the 124 respondents who participated in my survey. Your feedback has been indispensable in understanding user requirements and validating the requirements for my application.

Lastly, I would like to thank my family and friends for their countless support and encouragement over the course of this challenge.

Abstract

Physical inactivity is an ongoing public health problem, as there are a high number of people who quit exercising, 45-80% of people will quit exercising within the first couple of years of starting a program. Most current fitness applications do not provide users with long-term engagement because they have one-size-fits-all workout programs, static reminders and lack personalization. The goal of the project is to design and develop a Smart Workout Planner & Tracker mobile application that addresses exercise adherence issues for university-aged individuals 18-40 years of age.

A comprehensive literature review of 20 academic papers analyzed the theoretical framework and barriers to exercise adherence, effectiveness of digital interventions, Self-Determination Theory, algorithmic generation of workouts, Just-In-Time Adaptive Interventions (JITAI) and gamification strategies. An online survey was conducted on 101 respondents, gathering requirements for the application, and identified that the main barriers to physical activity were lack of time (55.6%), lack of motivation (55.6%) and fatigue (58.1%). The highest ranked fitness application feature requested was the ability to track progress (87.1%), followed by having personalized workout plans (76.6%).

The Smart Workout Planner & Tracker will have three components; (1) algorithmic generation of workout plans based on user fitness level, fitness goals, available equipment and time restrictions; (2) adaptive scheduling with evidence-based optimal time to send reminders and; (3) gamification features such as achievement badges, streaks and visual representation of progress. The system architecture will utilize Flutter for cross-platform development and Firebase for the backend.

The entire system has been designed in this FYP1 report. It includes the use case diagrams, class diagrams, sequence diagrams and interface mock-ups. A 14 week timeline for implementation has been made for FYP2 with a goal of having it completed by July 2026. The aim of this project is to demonstrate an operational prototype to improve fitness feature-specific adherence through the development of a personalized and evidence-based intervention design.

Keywords: fitness application, exercise adherence, personalization, gamification, mobile health, Flutter.

Table of Contents

Contents

Copyright	iii
Declaration	iv
Acknowledgements	v
Abstract	vi
Table of Contents.....	viii
List of Tables.....	xii
List of Figures	xv
List of Abbreviations/Symbols.....	xviii
List of Appendices	xx
Chapter 1: Introduction	1
1.1 Overview	1
1.2 Problem Statement	1
1.3 Project Objectives	3
1.4 Project Scope.....	3
1.5 Project Limitations.....	4
1.6 Methodology	4
1.7 Target Audience	5
1.8 Summary.....	5
Chapter 2: Literature Review	6
2.1 Overview	6
2.2 The Exercise Adherence Problem.....	6
2.2.1 Adherence Rates and Dropout Statistics	6
2.2.2 Barriers to Exercise Adherence	7
2.2.3 Determinants of Adherence	7
2.3 Digital Fitness Interventions	8
2.3.1 Promise of mHealth Applications	8
2.3.2 User Engagement and Abandonment.....	8
2.3.3 Limitations of Current Fitness Apps	9

2.4 Theoretical Foundations for Behavior Change	9
2.4.1 Self-Determination Theory (SDT)	10
2.4.2 COM-B Model and Theoretical Domains Framework	10
2.4.3 Behavior Change Techniques	11
2.5 Algorithmic Approaches to Workout Plan Generation	11
2.5.1 Integer Linear Programming (ILP)	12
2.5.2 Genetic Algorithms	13
2.5.3 Implications for Application Design	14
2.6 Just-in-Time Adaptive Interventions (JITAI)	14
2.6.1 JITAI Framework and Core Features	14
2.6.2 Reminder Optimization Using Reinforcement Learning	15
2.6.3 Personalized Intervention Criteria	15
2.7 Gamification and Motivational Features	16
2.7.1 Gamification Elements in Fitness Apps	16
2.7.2 Long-Term Engagement Strategies	17
2.8 Comparative Analysis	17
2.8.1 Identified Gaps in Current Fitness Applications	18
2.8.2 Feature Comparison Matrix	18
2.8.3 Analysis of Comparison Results	19
2.9 Summary	21
Chapter 3: Requirement Analysis	22
3.1 Overview	22
3.2 Fact-Finding Techniques	22
3.2.1 Justification	22
3.2.2 Questionnaire Design	23
3.3 Survey Results and Analysis	24
3.3.1 Demographics	24
3.3.2 Current Exercise Habits	26
3.3.3 Fitness App Experience	27
3.3.4 Barriers to Exercise	29
3.3.5 Feature Importance	30
3.3.6 Workout Preferences	35

3.3.7 App Preferences	38
3.3.8 Open Feedback	40
3.4 Requirements Specification	41
3.4.1 Functional Requirements.....	41
3.4.2 Non-Functional Requirements	42
3.4.3 User Requirements.....	43
3.5 Summary	44
Chapter 4: System Design	46
4.1 Overview	46
4.2 Architecture Diagram.....	46
4.3 Use Case Diagram.....	49
4.4 Activity Diagram	52
4.5 Class Diagram	57
4.6 Entity Relationship Diagram	59
4.7 Sequence Diagram	61
4.7.1 User Registration & Login.....	61
4.7.2 Generate Workout Plan	64
4.7.3 Complete Workout Session	66
4.8 Interface Design	69
4.9 Summary.....	72
Chapter 5: Implementation Plan	73
5.1 Development Phase (Front-end and back-end coding).....	73
5.2 Testing Phase.....	76
5.2.1 Unit Testing.....	76
5.2.2 Integration Testing	77
5.2.3 System Testing	78
5.2.4 User Acceptance Testing (UAT)	80
5.3 Deployment Phase.....	82
5.3.1 Deployment Stages	82
5.3.2 Deployment Platform	83
5.3.3 Alpha Release	83
5.3.4 Beta Release	84

5.3.5 Final Release.....	84
5.3.6 Deployment Checklist.....	85
5.5 Summary.....	85
Chapter 6: Conclusion	86
6.1 Summary.....	86
6.2 Expected Contributions	87
6.3 FYP2 Plan.....	87
References	xii
Appendix A	xiv
Appendix B	lvi

List of Tables

Table	Title
Table 2.1	Feature Comparison of Fitness Applications
Table 3.1	Survey Sections
Table 3.2	Functional Requirements Derived From Survey Responses
Table 3.3	Non-Functional Requirements Derived From Survey Responses
Table 3.4	User Requirements Derived From Survey Responses
Table 3.5	Key Findings From Survey
Table 4.1	Frontend Layer of Architecture Diagram
Table 4.2	Backend Layer of Architecture Diagram
Table 4.3	External API's Layer of Architecture Diagram
Table 4.5	Phases of Workout Plan Generation Process
Table 4.6	Description of Classes From their Categories
Table 4.7	Description of Entities From Their Categories
Table 4.8	Flow of Parties to Complete the Registration Flow
Table 4.9	Flow of Parties to Complete the Login Flow
Table 4.9.0	Flow of How Data is Retrieved

Table 4.9.1	Flow of How Filtering & Configuration is Done
Table 4.9.2	Flow of Optimizing a Workout Plan
Table 4.9.3	Flow of Validating a Workout Plan
Table 4.9.4	Flow of Starting a Session
Table 4.9.5	Flow of Executing an Exercise
Table 4.9.6	Flow of Completing a Session
Table 4.9.7	Flow of Rating a Workout's Difficulty
Table 4.9.8	Flow of Achievement Checking
Table 4.9.9	Description of Each Screens From Mockup Interface
Table 5.1	Backend Development Process
Table 5.2	Frontend Development Process
Table 5.3	Integration Process
Table 5.4	Unit Testing List
Table 5.5	Integration Testing List
Table 5.6	Functional Testing Checklist
Table 5.7	Non-Functional Testing List
Table 5.8	UAT Procedure List

Table 5.9	SUS Questionnaire List
Table 5.9.1	SUS Score Interpretation
Table 5.9.2	Deployment Stages
Table 5.9.3	Deployment Platforms
Table 5.9.4	Alpha Release Activity List
Table 5.9.5	Beta Release Activity List
Table 5.9.6	Final Release Activity List
Table 5.9.7	Deployment Checklist

List of Figures

Figure	Title
1.0	Exercise Adherence Rates to Prescribed Programs
3.1	Google Form Questionnaire to Gather User Requirements
3.2	Age Distribution of Respondents
3.3	Gender Distribution of Respondents
3.4	Status Distribution of Respondents
3.5	Fitness Level Distribution
3.6	Exercise Frequency per Week
3.7	Exercise Location of Respondents
3.8	Fitness App Previously Used by Respondents
3.9	Reasons for Abandoning Fitness Apps
3.10	Barriers to Regular Exercise (Multiple Selection)
3.11	Most Valuable Solution an App Could Provide
3.12	Feature Importance Ratings - Mean Scores (1-5 Scale)
3.13	Horizontal Bar Chart for Feature Importance Ranking
3.14	Primary Fitness Goal of Respondents

3.15	Preferred Workout Days per Week
3.16	Preferred Workout Duration per Session
3.17	Equipment Access (Multiple Selection)
3.18	Preferred Time of Day for Workouts
3.19	Preferred Method When Unsatisfied with Plan
3.20	Helpfulness of Suggested Starting Weights
3.21	Workout Reminder Preferences
3.22	Factors That Would Motivate Consistent Workouts
4.1	Architecture Diagram
4.2	Use Case Diagram
4.3	Activity Diagram - User Onboarding
4.4	Activity Diagram - Generate Workout Plan
4.5	Class Diagram
4.6	Entity Relationship Diagram
4.7	Sequence Diagram - User Registration & Login
4.8	Sequence Diagram - Generate Workout Plan
4.9	Sequence Diagram - Complete Workout Session

4.11	Interface Design Mockups for Login, Setting Profile and Goals, Home Dashboard, View Workout Plan
4.12	Interface Design Mockups for Active Workouts, Exercise Details, Tracking Progress, and View Achievements
5.1	Smart Workout Planner & Tracker Project Timeline (Gantt Chart)

List of Abbreviations/Symbols

Abbreviation	Definition
API	Application Programming Interface
BCT	Behavior Change Technique
BMI	Body Mass Index
BPNT	Basic Psychological Needs Theory
CET	Cognitive Evaluation Theory
COM-B	Capability, Opportunity, Motivation - Behavior
CRUD	Create, Read, Update, Delete
DBCI	Digital Behavior Change Intervention
ER	Entity Relationship
FYP	Final Year Project
ILP	Integer Linear Programming
JITAI	Just-in-Time Adaptive Intervention
MET	Metabolic Equivalent of Task
mHealth	Mobile Health

OIT	Organismic Integration Theory
PA	Physical Activity
RCT	Randomized Controlled Trial
RL	Reinforcement Learning
SDT	Self-Determination Theory
SUS	System Usability Scale
TDF	Theoretical Domains Framework
TDEE	Total Daily Energy Expenditure
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
WHO	World Health Organization

List of Appendices

Appendix	Title
A	Meeting Logs
B	Turnitin Similarity Index Page

Chapter 1: Introduction

1.1 Overview

In an era of digital innovation where information is lightning speed, awareness of staying healthy exponentially increases, which causes the arise of mobile fitness applications as a type of solution. These applications utilises advanced technologies like artificial intelligence, machine learning, and data analytics to come up with an application that offers personalised workout guidance, track progress, and increase motivation of users to reach their respective fitness goals. However, despite the abundance in fitness app being developed and marketed in each netizen's mobile application stores, a large gap remains between what these technologies were expected to bring, with individuals who failed to adhere to maintaining a healthy lifestyle via fitness applications.

This project aims to develop a smart, personalised fitness application designed to solve this problem by integrating a workout plan generated via algorithms with scheduling, real-time progress tracking, and adaptive motivational features. Compared to generic fitness apps, this project will tailor exercises to each users' constraints, such as time availability, equipment access, level of fitness, followed by personal goals. Moreover, this application combats existing problems by showcasing evidence-based behavioral strategies to increase discipline of users to keep engaging with the application.

By gathering insights from recent studies in digital health, human-computer interaction, and behavioral science, this project aims to create a practical, for-the-user centred solutions that help users to build sustainable fitness habits, alongside conducting frequent exercises in a disciplined manner.

1.2 Problem Statement

Regular physical activity is vital in keeping a healthy body and mentality. However, it is baffling to learn that the global adherence to exercising is concerningly low. Studies show that approximate 50% of netizens is not disciplined enough to be in an exercise program. Furthermore, the dropout rates reached 48-80% within just the first few years (Mahmood et al., 2023). This problem spreads among busy adults, students, and professionals who claimed to lack the time, having irregular schedules, and not having enough motivation, which became the main barrier to being consistent in exercising (Haroon & Rabia, 2025; Yang & Koenigstorfer, 2020).

Exercise Adherence Rates to Prescribed Programs



Figure 1.0: Exercise Adherence Rates to Prescribed Programs (Sabate, 2003; as cited in Mahmood et al., 2023)

Consequences of Physical Inactivity: Consequences of lacking physical activity can be severe. Such lifestyles have become the fourth leading factor of death in the world, since it became a factor in getting chronic conditions such as obesity, cardiovascular disease, diabetes, cancer, and mental health disorders (WHO, 2019; Pérez-Jorge et al., 2024). Researches shows that lack of exercise carries risks that can cause fatality, comparable to smoking (Park, 2012), with such lifestyle over 20 years, can double to risk (Moholdt, 2019).

Benefits of Regular Exercise: In contrast, regular physical activity offers substantial health benefits. Consistently, studies have revealed that regularly exercising assists in preventing noncommunicable diseases, which increase resilience against mental illness and reduce cognitive decline (Brown et al., 2024; Warburton et al., 2006). In addition, physical activity enhances psychological well-being by reducing stress and depression, increases one's mood, and getting better self-esteem (Pérez-Jorge et al., 2024). These benefits highlights the vitality of developing effective interventions that promote sustained exercise adherence.

While digital fitness applications is known to improve short-term adherence (Lang et al., 2022), a majority of them fails to provide truly personalised, adaptive, and context-aware guidance. Most of these applications rely on static workout plan templates, simple and generic reminders, and simplistic tracking mechanisms that do not consider the dynamic nature of precious lives, preferences, and progress. Moreover, long-term engagement remains a major challenge. This is because users often disengage when workouts become repetitive, irrelevant, or overly demanding (Du & Deng, 2025; Zhu et al., 2024).

Given these significant health implications, there is a clear need for a fitness application which:

1. Dynamically personalises workout plans, based from user's constraints (i.e. time, equipment, goals).
2. Integrates scheduling and adaptive reminders to align with users' daily routines.
3. Offers evidence-based behavioral techniques, such as setting goals, self-monitoring, and gamified responses in order to sustain motivation.
4. Provides a seamless, intuitive user experience which reduce friction and encourages habitual use.

This project covers all of these aspects by developing an application that generates user tailored workouts that adapts in real-time to user feedback, progress, and contextual factors.

1.3 Project Objectives

The project aims to accomplish the following goals:

1. Design and develop a working mobile fitness application that provides users with customized workout plans based on algorithmic optimisation techniques (e.g., Integer Linear Programming and Genetic Algorithms).
2. Establish a user-friendly scheduling system that will accommodate the user's availability and integrate into their daily routines.
3. Create and incorporate a set of core tracking and motivational features, which will include:
 - A calendar-based workout log;
 - Achievement badges and visual representation of progress;
 - A countdown timer to guide the user through their workouts.

1.4 Project Scope

This project consists of four major parts:

1. Front-End Development: A mobile application available for Android (with an iOS version potentially to follow) built utilizing a modern framework (e.g., React Native, Flutter).
2. Back-End System: A cloud-based service that will be used for authentication of users, data storage, workout generation and potentially utilize Firebase or could create a custom-built Node.js/Python API.
3. Workout Engine: Implementing optimization models (Kerr, 2025; Eriksson, et al.) to develop personalized workout plans based on user-entered data.

Feature Set:

1. User onboarding (fitness assessment, setting goals and determining what equipment they have access to);
2. Scheduling workouts (daily/weekly) with reminders (adaptive to latest completed workout);
3. Exercise library (including video/text instructions);
4. Dashboard (showing calendar of completed workouts, and progress relative to stated goals);
5. Rest timer between sets and for total time of completed workout.

Testing and Evaluation: Conducting usability tests with a select group of target-end users, during which feedback will be collected regarding the overall interface design, functional design and perception of motivation..

Out of Scope:

- Biometric integration (wearable devices (HRM) for heart rate);
- Social (leader-board, community challenges);
- Multi-language support or regionally focused customizations;
- Commercially deployed or published on any app stores.

1.5 Project Limitations

The following limitations apply to the complete creation and assessment of this project:

1. Technical Limitations: This application will be developed as a prototype, with the most critical functions included, but the finished product may include polished and production-ready functions working correctly.
2. Platform Limitations: Development will focus on Android first and later the application will need to be developed for a compatible iOS version.
3. Simplicity of Algorithms: The workout generation algorithm will utilize predefined rules and optimization heuristics rather than more sophisticated machine learning algorithms that rely on many data points to be generated.
4. Limited User Testing: The usability portion of testing will only be conducted with a limited number of users due to time and resource constraints ($\approx 10-15$), which will limit the ability to generalize the results of the findings.
5. User Data Security: User data will be stored in a secure manner, however, the application will not undergo formal security certification or auditing compliance with security standards such as HIPAA or GDPR.

1.6 Methodology

The project will implement an Agile-inspired iterative development approach, consisting of four phases:

1. Phase 1: Define the functional and non-functional requirements by reviewing existing literature (e.g. papers written by Du & Deng; Zhu et al; Mahmood et al).
2. Phase 2: Develop wireframes (for the user interface), database schema and system architecture diagrams.
3. Phase 3: Build the modules for the front-end of the system (e.g. views), back-end (e.g. server-side code) and algorithms (e.g. business rules).
4. Phase 4: Perform unit and integration testing during the development of the modules where we:
 - Perform UAT with actual users in order to assess how usable and engaging the system is using the SUS and qualitative feedback.
 - User acceptance testing (UAT) with target users to evaluate usability and engagement using the System Usability Scale (SUS) and qualitative feedback.
5. Phase 5: Analyze the results of the user evaluation/testing and make iterative improvements to the system.

1.7 Target Audience

The audience for this app are:

- Busy professionals or students aged 18-40 that struggle to find time for structured workouts
- Anyone new to fitness or an intermediate who would like to have a personalized workout plan without the cost of hiring a personal trainer.
- People that don't have access to lots of equipment like people that work out at home (or have limited options) who need a tailored workout routine.
- People that are motivated to track their progress digitally and like to reach fitness goals but aren't happy with generic fitness apps.

This matches up with the users in the market for fitness app research in the past 12 months (Haroon & Rabia, 2020-2021; Perez-Jorge et al.,2021) and also shows the need for more personalized and engaging digital fitness solutions.

1.8 Summary

This chapter has described the motivation for the development of a personalized fitness app to help bridge an important gap in existing digital fitness applications by providing enhanced opportunities for users to exercise more frequently and have greater interaction due to algorithmic personalization, intelligent schedule generation, and evidence-based motivational design. Subsequent chapters will discuss the literature review, system design, implementation, testing and conclusions regarding this project.

Chapter 2: Literature Review

2.1 Overview

Physical inactivity is a health issue of concern on the international level, and its effects on chronic disease prevention and overall health are considerable. Although most people have been educated on the advantages of engaging in regular physical activities, the problem is still there with the adherence percentage of most people to physical activity guidelines being low. The presented literature review is a synthesis of 20 scholarly articles exploring the use of digital interventions, especially mobile fitness applications, in the context of overcoming the challenge of exercise adherence using personalized and evidence-based strategies.

The literature reviewed belongs to various fields such as exercise science, behavioral psychology, human-computer interaction, and artificial intelligence, which makes effective fitness apps design interdisciplinary in nature. An overall story can be devised using all these papers: defining the problem, theoretical backgrounds, and practical implementation strategies and empirical validation.

This chapter has the following structure: Section 2.2 discusses the problem of exercise adherence and barriers to it. Section 2.3 addresses the issue of digital fitness interventions, their promise and pitfalls. Section 2.4 discusses theoretical premises of behavior change. In section 2.5, algorithmic methods to workout plan generation are given. The just-in-time adaptive interventions are discussed in section 2.6. Section 2.7 explores the concept of gamification and motivation. Section 2.8 gives a comparative analysis of the same and a summary of the main findings is given in Section 2.9.

2.2 The Exercise Adherence Problem

Based on the literature reviewed, exercise maintenance can be viewed as complex and difficult to resolve due to its many dimensions. To create effective digital solutions, we need to understand the barriers, determinants, and extent of exercise maintenance issues.

2.2.1 Adherence Rates and Dropout Statistics

According to Mahmood et al., (2023) there is no single best way of measuring adherence and the reported rate of adherence to home-based exercise is only 24-40%. Furthermore, the rate of long-term adherence (in the 1st year of an exercise program) is 30-50% and drop-out rates for exercise programs are reported to be between 45-80% in the first 4 years of being completed (Ice, 1985, cited in Mahmood, et al).

These statistics suggest that adherence to exercise is an issue for many people. Even though people start exercise programs with the best intentions, the large majority of them will not be able to maintain a regular exercise routine over time. The challenges that have prevented these people from completing their exercise programs mature have been observed across many different groups of individuals, with different types of exercise and use various types of interventions, indicating that for many an individual's adherence is a substantial barrier to maintaining an active lifestyle and for most individuals require some sort of comprehensive approach.

2.2.2 Barriers to Exercise Adherence

Time Constraints Are The Most Frequently Identified Barrier To Participation In Physical Activity Among University Students; 90% Of Studies Reviewed (39 Total) Found This To Be A Significant Barrier (Brown Et Al., 2024) And Haroon (2025) Also Found It To Be A Key Barrier In Finland. While These Time Barriers Are Usually Due To Competing Priorities Or Poor Time Management And Are Not An Indication Of Absolute Scarcity, It Is Likely That Interventions...Assisting Users With Their Priorities And Committing Time To Exercise Would Be Successful At Overcoming Time Barriers.

Other Frequently Identified Barriers Include:

1. Demotivation
2. Poor access to equipment or facilities.
3. Financial Constraints
4. .Lack Of Social Support
5. Limited Health Literacy
6. Controlling life demands (work, family, studies).

According to Mahmood et al. (2023), non-adherence is also caused by lack of primary care providers, poor communication with healthcare providers and cognitive deficits.

2.2.3 Determinants of Adherence

Mahmood et al. (2023) identified five interconnected domains that influence adherence:

1. Personal factors: Motivation, self-efficacy, knowledge, and prior exercise experience.
2. Disease-related factors: Symptoms, severity of illness, and perception of illness seriousness.
3. Therapy-related factors: Complexity of therapy, length of therapy, and changes to behaviour required.

4. Provider-related factors: Quality of communication with providers, level of provider support, and quality of the therapeutic relationship.
5. Health system factors: Accessibility to health services, cost of care, and availability of physical health resources.

Because of the multidimensional nature of these factors, it is clear that effective interventions should involve addressing multiple barriers at once, not targeting one factor at a time. Brown et al. (2024) used the COM-B model to identify 56 unique barriers and facilitators that can be grouped into 12 theoretical domains, further supporting the need to take into consideration all dimensions when trying to understand exercise behaviour.

2.3 Digital Fitness Interventions

With the rapid rise of mobile health (mHealth) applications, there are now opportunities to deliver customized exercise programs at scale. Understanding what digital health solutions can do to help improve people's ability to be active as well as how they may be harmful is necessary in order to take full advantage of this new technology.

2.3.1 Promise of mHealth Applications

According to Yang and Koenigstorfer (2020), the use of physical activity apps can reduce the decline in physical activity. For example, their longitudinal study demonstrated that individuals who used an activity app were less likely to experience a decrease in physical activity during COVID-19 lockdowns than those who did not use the app. Therefore, the findings from this research support the value of using digital fitness applications as a means of preserving physical activity during stressful times.

Lang et al. (2022) provide a synthesis of RCTs evaluating the effects of using digital technologies on exercise adherence. Ultimately, 70% of the 10 trials showed that digital technologies positively influence exercise adherence. As a result of their findings, digital technologies could potentially enhance the likelihood of exercising regularly when added to a traditional home exercise program. Examples of benefits associated with digital technologies include but are not limited to the following:

- Accessible and convenient
- Provide real-time feedback and allow for tracking
- Provide personalized content to users
- Cost-effective compared to working with a personal trainer
- Can reach large populations

2.3.2 User Engagement and Abandonment

The promise of digital interventions is offset by troubling statistics about abandonment rates. According to the work done by Alberts et al. (2024), 53% of mHealth apps are

uninstalled within 30 days of being downloaded. This indicates that just because someone has adopted an app doesn't mean they will stay engaged with it.

As identified by Mustafa et al. (2022), the main reason people abandon mobile health apps includes:

- Amotivation 31.6%
- Did not have features they wanted 18.7%;
- Did not enjoy using it 10.0%
- Usability issues 8.6%

Similarly, Kim and Lee (2023) conducted a text mining project analyzing 100,000 reviews of Fitbits and identified complaints related to functionality, compatibility, and being charged for services as key factors in driving people away from these device

2.3.3 Limitations of Current Fitness Apps

The design of the digital intervention plays a crucial role in determining the effectiveness of the intervention in terms of maintaining a user's long term engagement. As reported by Du and Deng (2025), exercisers expect their workout to be personalized and to have their workouts adjust to their information and progress tracking and their efforts documented clearly. However, most commercially available exercise applications do not provide these personalized results for exercise participants and use the following:

- Static workout templates
- Generic reminders without context-awareness
- Basic tracking capabilities
- Assumed to be for everyone

The primary challenge of motivating the user with the exercise application lies not only in the development of a technology-based solution to improve use after adoption but rather in a user's motivation to continue using the exercise application after initial satisfaction from the purchase of the application and their ability to learn the value of the change from their behavior.

2.4 Theoretical Foundations for Behavior Change

To create effective designs for an exercise application, it is recommended that design organizations ground their design in well-established behavior-change theories. The literature reviewed represents complementary theoretical frameworks that can provide

complementary resources for use in a comprehensive resource for designing an intervention.

2.4.1 Self-Determination Theory (SDT)

In their study of Self-Determination Theory, founded by Decy and Ryan (2000), Alberts et al. (2024) review how this theory explains the maintenance of motivation through the fulfillment of three basic psychological needs: Autonomy –

1. **Autonomy:** the feeling of having choice and control over one's own behaviours.
2. **Competence:** feeling effective and competent in achieving goals
3. **Relatedness:** feeling close to others and having a sense of belonging.

Importantly, they identify a common design flaw with the majority of behaviour change techniques (BCTs) within fitness apps, which focus on using the components of SDT to create engagement with the technology, rather than helping users to internalise the value of changing their behaviour. This distinction is extremely significant, as BCTs that make exercise "fun" only through the use of gamification, may experience a loss of appeal once the novelty has worn off, while those that help their users to internalise the reasons for exercising may be able to help their users develop behaviours and achieve change over the long-term.

SDT distinguishes between extrinsic motivation (driven by external factors to reward), and intrinsic motivation (driven by an internal factor of enjoyment or satisfaction). Effective interventions will assist the user in making the transition from extrinsic to intrinsic motivation to increase the likelihood of long-term behaviour change.

2.4.2 COM-B Model and Theoretical Domains Framework

Brown et al. (2024) have utilized the COM-B model and Theoretical Domains Framework (TDF) to develop a framework for mapping the determinants of behaviour in the following three areas:

- **Capability:** The physical and psychological ability to perform a behaviour.
- **Opportunity:** Includes the physical and social environment that creates a barrier to performing a behaviour.
- **Motivation:** Refers to both reflective (conscious) and automatic (habitual) processes that influence one's motivation to perform a behaviour.

Brown et al. (2024) identified 56 barriers and facilitators through the TDF framework across 12 domains of TDF. Therefore, the approach is comprehensive in representing factors that influence exercise behaviour. According to the COM-B model, behaviour change intervention strategies need to specifically address deficits in all 3 domains, as

improving only capability will not lead to success if other barriers exist in motivation or opportunity

2.4.3 Behavior Change Techniques

hu et al. provide an overview of over a hundred behavioral change strategies that have been developed for use in digital interventions. Here are some of the most frequently employed techniques;

- **Self-monitoring of behavior:** Keeping track of workouts, steps taken, progress made;
- **Goal-setting:** Developing clearly defined and measurable goals;
- **Prompts and cues:** Providing reminders to initiate exercises;
- **Feedback on behavior:** Providing information regarding performance and achievement;
- **Social support:** Connecting with others;

Regarding forming a habit: The techniques used are primarily focused upon helping establish motivation, creating cue-behavioural associations and providing reinforcement for positive behaviour.

Mahmood et al. support the notion of including multiple strategies in interventions used to support compliance. These include motivation enhancement, self-efficacy improvement, and cognitive behavioural theory. Each of these theories is similar in that; to achieve accurate behavioural change, one must address all factors contributing to the change simultaneously. Further, long-term change will occur only if there is an intrinsic motivation versus relying on extrinsic incentives.

2.5 Algorithmic Approaches to Workout Plan Generation

Heuristics, professional intuition, and rigid templates that don't allow for variation in application across all different types of users, are all the base of how traditional workout plans are made (Kerr, 2025). Traditional methods also don't typically account for individual limits like time to train, access to equipment, and how long they need to recover, along with other personal goals for each person's level of fitness. In contrast to traditional methods, mathematical optimization can be used to create a systematic alternative by defining workout design as a structured decision-making problem; thus creating an effective personalized workout plan that is adaptable to meet all individuals' needs.

2.5.1 Integer Linear Programming (ILP)

Kerr (2025) presents a new method for generating customized training programs automatically using linear programming (LP) methods called Integer Linear Programming (ILP). This type of optimization considers an objective function (to max or min) with constraints based on linear relationships (no fractional variables). ILPs have been successfully used in operation-research fields. Examples include scheduling and resource allocation (Winston & Goldberg 04).

The ILP model for workout programming utilizes various components of the application.

The ILP model for workout planning operates on several core components:

Sets and Ranges:

- Let E be the set of all exercises, indexed by $i \in \{1, 2, \dots, N\}$
- Let D be the set of workout days, indexed by $j \in \{1, 2, \dots, \text{day_count}\}$
- Let M be the set of target muscle groups, indexed by m

Parameters:

- $s_i \in [1, 10]$: Effectiveness score for exercise i
- $\text{type}_i \in \{\text{compound}, \text{isolation}\}$: Exercise classification
- $\text{tim} = 1$ if exercise i targets muscle m , else 0
- $\text{focus}_j \subseteq M$: Target muscle groups for day j
- max_compounds_j : Maximum compound exercises allowed per day

Decision Variable:

$$x_{i,j} = \begin{cases} 1, & \text{if exercise } i \text{ is assigned to day } j \\ 0, & \text{otherwise} \end{cases}$$

Objective Function:

The total overall training effectiveness per session is maximized by the model through all available exercises:

$$\text{maximize } z = \sum_{i \in E} \sum_{j \in D} s_i \cdot x_{i,j}$$

Constraints:

Muscle Coverage — Each target muscle group must be trained at least once per day:

$$\sum_{i \in E} t_{im} \cdot x_{i,j} \geq 1 \quad \forall j \in D, \forall m \in \text{focus}_j$$

Compound Lift Limit — Prevents overloading by capping taxing compound movements:

$$\sum_{i \in E} x_{i,j} \cdot \mathbb{I}(\text{type}_i = \text{compound}) \leq \text{max_compounds}_j$$

The model selects from multiple types of exercises per user-specific constraints to maximize workout effectiveness while guaranteeing feasibility and optimality (Bertsimas & Tsitsiklis, 1997).

2.5.2 Genetic Algorithms

Eriksson et al. (2021) provide a different way of developing an athlete-specific training program that emulates professionally coached workouts through the use of a genetic algorithm combined with machine learning. Their developmental methodology developed is called GERT (Genetic and Random Trees training planner) and demonstrates that an expert coaching methodology is capable of being algorithmically captured.

The functionality of GERT is accomplished in 2 stages:

Stage 1 — Learning Coach Style:

- Create a regression model to define the distribution of training loads across multiple training sessions
- Determine which training session types occur in which order by utilizing historical plans

Stage 2 — Development of Training Programs: Through use of a genetic algorithm, iteratively improve a population of training session plans by:

- An initial sample of session type workouts is drawn randomly from a database
- Retain the sessions that provide the best experience for the athlete being evaluated.
- Combine the successful training session workouts with each other.
- Add different variations to the sessions to explore new workout plans.
- Evaluate the candidate plans using a multi-criteria loss function that is based on five metrics.

The loss function evaluates plans based on five metrics:

1. eD : Training load distribution was similar among sessions in the program.
2. eZ : Training load distribution was similar among training intensity zones.
3. eT : The order of session types was similar.

4. *eS*: The number of sessions completed of each session type was relatively equal.
5. *eF*: The sessions are suitable for the specialties of the athlete.

The results of GERT show a perfect $R^2 = 1.00$ for the distribution of training loads, thus demonstrating that the use of GERT to develop an athlete-specific training program provides an almost identical representation of a professional coach's workout planning process when compared to a baseline nearest-neighbor plan ($R^2 = 0.88$).

2.5.3 Implications for Application Design

Both methods can give you reasonable ways to use intelligent workout generation in mobile fitness apps. ILP provides optimal results (guaranteed) and requires less data to work, which makes it a good choice for an application where the only driver of personalization would be the user's input. On the other hand, genetic algorithms require more computing power but can produce results that match the personal coaching style of an expert coach if historical training data is available.

Both approaches are supported using research on user experience. Du and Deng (2025) found that users of the fitness app want their workout plan to be personalized based on their information, want to see their progress tracked, and want the results to be displayed in an easy-to-understand format. Therefore, it may be necessary to incorporate optimization algorithms into user interfaces so that personalization can be achieved transparently to users and be within their control.

2.6 Just-in-Time Adaptive Interventions (JITAI)

Beyond planning what users should do, the literature examines how to optimally deliver interventions at the right moment and in the right context.

2.6.1 JITAI Framework and Core Features

According to Wunsch et al. (2020), JITAI frameworks should typically examine five essential components of an adaptive intervention in order to be effective:

1. Address real-time needs: the intervention matches the user's present state.

2. Adapt using input data: the intervention's response to the user changes based upon either the sensor or the user data being collected.
3. System-triggered: the intervention is delivered automatically based upon detection of specific conditions
4. Goal-oriented: the intervention is developed based upon the user's desired outcome.
5. Customized to user preferences: the delivery of the intervention should also be tailored specifically to the user based upon their preferences.

The JITAI framework enables adaptive interventions that adaptively change in response to a user's individual context instead of providing fixed-schedule delivery.

2.6.2 Reminder Optimization Using Reinforcement Learning

Through the use of reinforcement learning (RL), Wang et al. (2021) put the JITAI principles into practice and studied their effects on timing of reminders optimally for users. The feasibility study showed that:

- 83.3% of RL-optimized reminder times generated user reaction within 50 minutes after sending.
- 66.7% of physical activity occurred within five hours of sending RL-optimized reminders.
- A pre-learned RL reminder model can be used to address the cold-start problem associated with new users of the RL-optimized reminder system.
- A maximum of 14 reminders can be sent each week to avoid user fatigue.

The results of this study provide evidence to show that using this technique for JITAI is a substantial improvement to traditional static reminder systems in that this method will learn and adjust its method for delivering reminders based on the individual characteristics of each user for all reminders sent.

2.6.3 Personalized Intervention Criteria

Ikegaya et al. (2025) present important evidence indicating that personalized intervention criteria are superior to uniform criteria. They found both personalized and uniform criteria resulted in substantial physical activity increases; however, personalized criteria, derived from initial individual variability in physical spatial and temporal behaviours, were found to better support users' needs and activity preference patterns both prior to and during the intervention period.

Kunkel et al. (2023)'s investigation of push notification strategy demonstrated that notifications that promote engagement regarding gamefulness can improve user engagement, however, the timing, frequency, and content of the notifications must be well-calibrated. Too many notifications cause dissatisfaction while too few notifications

can also lead to disengagement; therefore, effective design requires the establishment of an optimal balance point for each user.

Overall, the above findings support the theoretical rationale for utilizing personalized design elements when designing JITAIs and support that "one size fits all" interventions are not the best option.

2.7 Gamification and Motivational Features

Gamification—the application of game design elements in non-game contexts—has emerged as a popular strategy for enhancing user engagement in fitness applications.

2.7.1 Gamification Elements in Fitness Apps

Virtual rewards (such as points, badges, and trophies) are reported by Zhu et al. (2024) to be the most used gamification method with 39% of the total reviewed studies. The following is a list of other widespread gamification methods:

- Progress bars and achievement systems
- Leaderboards and social competition
- Streaks and consistency rewards
- Challenges and quests
- Customization of avatars

According to Pérez-Jorge et al. (2024), gamified physical activity has positive mental health effects in adolescent populations, including well-being, fun, socializing, satisfaction and motivation. Therefore, there are dual benefits of using gamification to promote both activity levels and psychological well-being.

Yang and Koenigstorfer (2020) found that gamification features were especially useful in mitigating declines in physical activity during COVID-19 lockdown periods. This implies that gamification can be particularly helpful to remain motivated through difficult times (e.g., during times of low motivation).

The literature warns against relying solely on extrinsic (external) rewards to maintain behaviors. Purely extrinsic forms of motivation as identified by Alberts et al. (2024) will likely not sustain behaviors as reward systems become established, or when users stop using mobile applications. Effective gamification should augment (or support) intrinsic (internal) motivation, rather than replace it.

2.7.2 Long-Term Engagement Strategies

There is an important uncertainty about the longevity or sustainability of the effects of digital interventions. In the 2022 paper by Lang et al., they provide good evidence for short-term effectiveness (four out of four studies show positive effect at less than or equal to six weeks); however, they find limited evidence of medium-term effectiveness (two out of four studies show positive effect between eight to twelve weeks). Therefore, while initial effectiveness may be achieved with an intervention, it does not provide assurance of sustained engagement.

Findings from Lunde et al. in 2020 are supporting evidence that, with appropriately designed application-based interventions, exercise capacity improvements achieved post-cardiac rehabilitation can be sustained over a 12-month period. This would suggest that it is possible to sustain engagement; however, it will require engaging intentional design for long-term use as opposed to purely through initial adoption.

The paper by Sun et al. in 2021 indicates personalization based upon individual differences improves outcomes with "feeling" personality types demonstrating an 18.15% higher level of adherence as compared to "thinking" personality types when matched for messaging. Therefore, strategies for long-term engagement should include specific individual differences as part of the personalization algorithms.

According to the paper by Mustafa et al., (2022) approximately 74.6% of users rated goal-setting as essential; however, only approximately 30.6% of users rated gamification features positively. As such, the prioritization of self-regulating core features as compared to gamification features (which will enhance but not provide foundation) is warranted.

As demonstrated by these collective findings, achieving sustained engagement requires the following:

1. Moving users from extrinsic to intrinsic motivation
2. Continuous adaptation to user needs and preferences
3. Personalization based on individual characteristics
4. Addressing the full range of COM-B determinants rather than motivation alone

2.8 Comparative Analysis

In this section, a comparative review of existing fitness apps will be made in relation to the new Smart Workout Planner & Tracker App. The comparison will focus on points that were previously established as being the most important features for addressing barriers related to compliance with exercise based on a literature review conducted prior to this comparison.

The four most frequently used fitness apps among the defined target population that were originally derived from the results of a survey included: Apple Fitness (25%), Samsung Health (12.9%), MyFitnessPal (11.3%) and Home Workout (10.5%).

2.8.1 Identified Gaps in Current Fitness Applications

According to the literature, there are a number of major issues with current fitness applications that have contributed to their high rate of abandonment. Alberts et al. (2024) cite a 53% uninstall rate for mHealth applications within 30 days of download, whereas Mustafa et al. (2022) have identified the main reasons for abandonment as the lack of motivation (31.6%), absence of required features (18.7%), lack of enjoyment (10%), and poor usability (8.6%).

Du & Deng (2025) specifically state that most commercially available applications do not meet user expectations for personalization but instead rely on static templates for workouts, generic reminders that are not aware of context, simple tracking methods, and a one size fits all approach. Similarly, Kim & Lee (2023) used text mining on 100,000 Fitbit reviews to corroborate the findings of the reviewers who mentioned complaints about paid services, compatibility issues, and functionality as being the top reasons for user dissatisfaction.

2.8.2 Feature Comparison Matrix

In Table 2.1, a summary of many popular fitness application features versus the Smart Workout Planner and Tracker are shown. The comparison used the same evidence-based features described in the Evidence-Based Features section in the first sections of this chapter.

Feature	MyFitness-Pal	Home Workout	Samsung Health	Nike Training	Proposed App
Personalized Plan Generation	X	Partial	X	Partial	✓
Adaptive Scheduling	X	X	X	X	✓

Smart Reminder Timing	X	X	X	X	✓
Progress Statistics	✓	✓	✓	✓	✓
Achievement Badges	✓	✓	✓	✓	✓
Equipment-Flexible Workouts	X	✓	X	Partial	✓
Video Tutorials	X	✓	X	✓	✓
Beginner-Focused Design	X	✓	X	Partial	✓
Free (No Paywall)	Partial	Partial	✓	Partial	✓
Rest Timer with Alarm	X	✓	X	✓	✓

Table 2.1: Feature Comparison of Fitness Applications

2.8.3 Analysis of Comparison Results

A comparison of current solutions shows several significant differences between what is currently available and what will be provided by the Smart Workout Planner & Tracker.

1. **Personalization Gap:** All the currently surveyed apps (Nike Training Club, Home Workout) offer some customizable features; however, none of them offer truly personalized workout programs based on user characteristics such as fitness level, exercise equipment available, amount of time available, and desired goals at the same time. MyFitnessPal and Samsung Health focus on tracking things, not providing workouts based on a user's needs. The new app addresses the personalization issue through algorithmically generated workout plans getting detailed information about the user (Du & Deng, 2025).

2. **Adaptive Scheduling Gap:** None of the currently surveyed apps have any kind of adaptive scheduling feature that would allow a user to change the scheduled time of their workouts based on how available they are during particular time periods. All the current apps have static schedules that the user sets and only modify as the user chooses to change their schedule. The new app has incorporated intelligent (adaptive) scheduling for the user's preferences and limitations, which addresses the issue of the time barrier in 90% of the studies reviewed by Brown et al. (2024).

3. **Smart Reminder Gap:** All other apps have basic reminder systems that only remind users at a predetermined time. Aguilera et al. (2024) demonstrated that reminders that used optimally designed reinforcement-learning techniques resulted in 83.3% of users responding within 50 minutes, much higher than with static reminders. The new app will use evidence-based strategies for reminders consistent with the Just-In-Time Adaptive Interventions (JITAI) principles (Wunsch et al., 2024).

4. **Equipment Flexibility Gap:** A majority of respondents (46.8%) reported having access only to bodyweight exercises (as per Chapter 3). However, existing applications such as MyFitnessPal and Samsung Health do not provide sufficient support for developing a workout plan without any equipment. The proposed app solves this issue by providing a way to create a workout plan based on the type of equipment available.

5. **Cost Barrier:** Mustafa et al. (2022) have indicated that users tend to discontinue using applications due to their high cost, and respondents indicated a desire for free or less costly alternatives. Most popular apps use a freemium model that charges a fee to access some features. The proposed app will be completely free to address this issue of accessibility to users.

Comparing the Smart Workout Planner & Tracker to current solutions shows that it includes three features not included in other offerings:

1. The ability to generate an algorithmic-generated workout based on the simultaneous consideration of multiple user-defined constraints
2. The ability to schedule workouts based on a user's irregular schedule and available time to work out
3. The ability to provide a user with evidence-based motivating features (e.g., visual representation of progress made towards achieving a goal and receiving badges or accomplishments for achievements) that are optimally timed.

These three features fulfil the main reasons for abandoning workouts stated by Mustafa et al. (2022) and provide for all of the personalization expectations identified in Du and Deng (2025).

2.9 Summary

In this chapter, 20 academic publications were studied to establish a solid research-based foundation for the Smart Workout Planner & Tracker application. Key findings from those studies are summarized below:

The exercise adherence rates problem: You want to improve your exercise adherence rates. Exercise adherence rates for exercise equipment in a home fitness setting are very low. The exercise adherence rate for exercise equipment in the home is only 24-40%. If you drop out of an exercise program, you have about a 45-80% chance of dropping out of that program, within four years. The most common barrier for people to adhere to their exercise program is lack of time. This was identified as the most common barrier for exercise adherence, in 90% of studies examined (Mahmood et al., 2023). Lack of motivation and fatigue were the next most common barriers to exercise adherence.

The potential for digital intervention: While digital fitness tools have shown to have a positive impact 7 out of 10 times in a clinical trial (Lang et al., 2022), 53% of fitness apps will be deleted (uninstalled) by their owner within 30 days after they download them (Alberts et al., 2024). The primary reasons people will delete an app include amotivation (31.6% of users), missing features (18.7% of users), and poor usability (8.6% of users) (Mustafa et al., 2022).

Strategies Found to be Beneficial:

- **Customization of Plans:** Users desire a plan based on their circumstances (not just a pre-defined template) (Du & Deng, 2025)
- **Smart Reminders:** Reminders designed using Reinforcement Learning achieve an average response rate of 83.3% compared to traditional static reminders (Aguilera et al., 2024)
- **Gamification:** Virtual rewards are found to be part of 39% of successful interventions; goal tracking is the largest driver for individuals to engage in an activity (Zhu et al., 2024)
- **Theoretical Grounding:** Self-Determination Theory (SDT) offers a model of the elements of motivation (e.g., autonomy, competence, connectedness) that are necessary for sustainable motivation

Gap Identified: The comparison shows that there is no application that includes all three of these features for enhancing adherence - personalized plan creation, adaptively scheduled plans, and evidence-based reminders - as all three of these features have been found supportive in passive use of behavior change technologies in terms of enhancing adherence per literature review.

The findings provide the basis for the requirements analysis in Chapter 3 and system design in Chapter 4.

Chapter 3: Requirement Analysis

3.1 Overview

This is a requirements analysis of the Smart Workout Planner & Tracker mobile application. This analysis aims to validate the overall feature set of the proposed app and ensures that those features are aligned with what the app's intended target audience, university students and young adults (ages 18-40), require to successfully adhere to their exercise programmes despite constraints on time, lack of motivation, or inconsistency in their daily schedule, each affecting them in different ways, thus resulting in their inability to exercise regularly.

3.2 Fact-Finding Techniques

3.2.1 Justification

There are many different methods to gather facts, such as interviews, observations, document reviews and surveys. The main Way to gather information on this project was through a survey.

Reasons for Choosing Survey:

- **Capability:** An online survey can provide a large group of individuals with valuable data in a short time. In a limited timeframe for this project, it is very important to get a large amount of information quickly.
- **Accessibility of Target Users:** The target population for this project, university students and young adults, ages 18 to 40, can easily be reached through online methods and social tools, therefore making it an easy way to distribute a survey.
- **Quantifiable Data:** Surveys provide a structured amount of data that can be used to validate requirements, as well as to prioritize characteristics of the proposed product according to how much importance to users attach to them, based on their feedback.
- **Anonymity:** When individuals can give their opinion without giving their name, individuals will provide more honest and useful feedback regarding their exercise habits and what either prevents them from exercising or what equipment they would choose, also known as social desirability bias.
- **Economic:** Surveys created on sites like Google Forms provide all of the information free of cost to create and distribute. Therefore, a survey is a good fit for an academic project that does not have a budget.

- **Standardization:** Every survey participant is asked the same questions, which provides consistency in data collection and allows for comparison of responses.

3.2.2 Questionnaire Design

An online survey was designed using Google Forms and distributed via social media platforms (WhatsApp, Instagram) and university networks.

Survey Structure:

- Total Questions: 40 questions across 8 sections
- Question Types: Multiple choice, checkboxes, Likert scale (1-5), and open-ended
- Estimated Completion Time: 10 minutes
- Distribution Period: February 3-4, 2026
- Target Respondents: University students and young working adults aged 18-40
- Total Responses Collected: 124

Survey Sections:

Section	Focus	Number of Questions
1	Demographics	4
2	Current Exercise Habits	4
3	Fitness App Experience	4
4	Barriers to Exercise	2
5	Feature Importance	13
6	Workout Preferences	6
7	App Preferences	5
8	Open Feedback	2

Table 3.1: Survey Sections

The survey questions were designed based on findings from the literature review (Chapter 2) to validate proposed features and understand user needs. Feature importance questions used a 5-point Likert scale to quantify user preferences for prioritization.

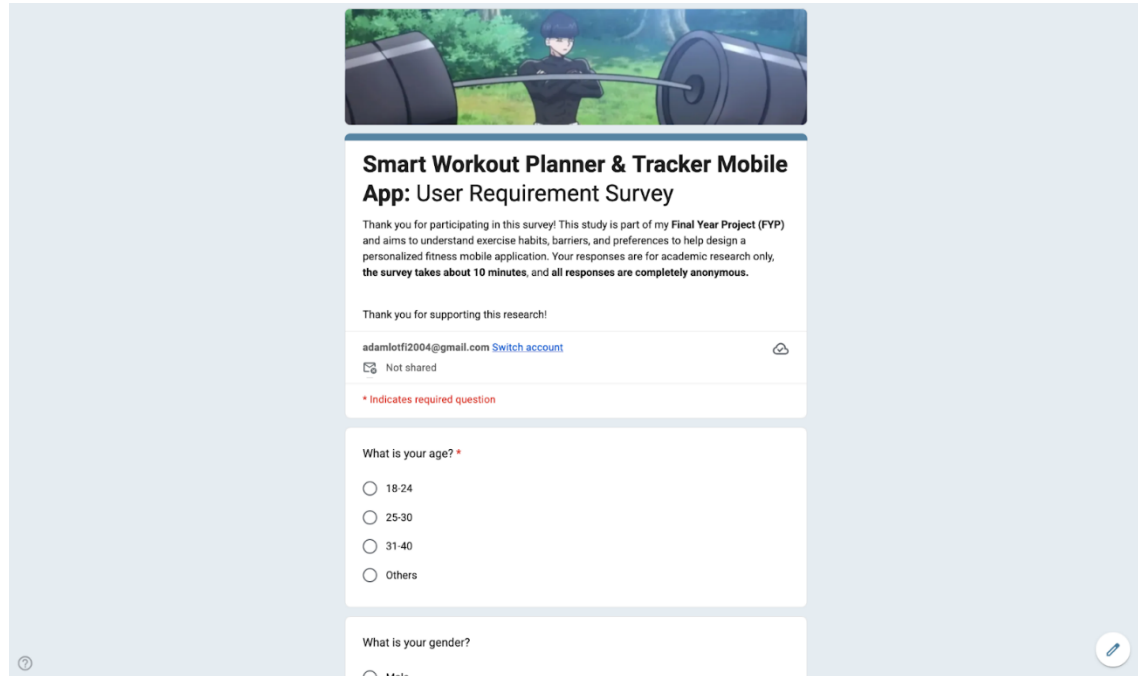
The image shows a Google Form titled "Smart Workout Planner & Tracker Mobile App: User Requirement Survey". The form has a header image of a person working out. The text inside the form says: "Thank you for participating in this survey! This study is part of my Final Year Project (FYP) and aims to understand exercise habits, barriers, and preferences to help design a personalized fitness mobile application. Your responses are for academic research only, the survey takes about 10 minutes, and all responses are completely anonymous." Below this, it says "Thank you for supporting this research!" and provides the email "adamlotfi2004@gmail.com" with a "Switch account" link. There is a "Not shared" status and a red asterisk indicating required questions. The first question is "What is your age?" with radio button options: "18-24", "25-30", "31-40", and "Others". The second question is "What is your gender?" with a radio button option for "Male".

Figure 3.1: Google Form Questionnaire to Gather User Requirements

3.3 Survey Results and Analysis

This section presents the analysis of 124 survey responses collected between February 3-4, 2026. The results are organized according to the survey sections and used to validate and refine the application requirements.

3.3.1 Demographics

The demographic profile of respondents was analyzed to confirm alignment with the target audience.

What is your age?

124 responses

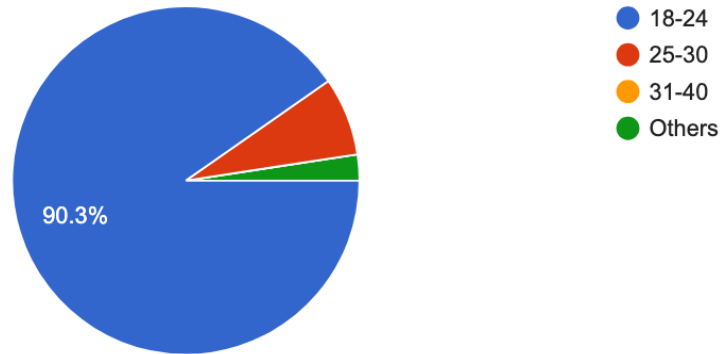


Figure 3.2: Age Distribution of Respondents

Age Distribution: As shown in Figure 3.2, the vast majority of respondents (91.1%) were aged 18-24, followed by 25-30 (5.9%), and others (3.0%). This strongly aligns with the target audience of university students and young adults.

What is your gender?

124 responses

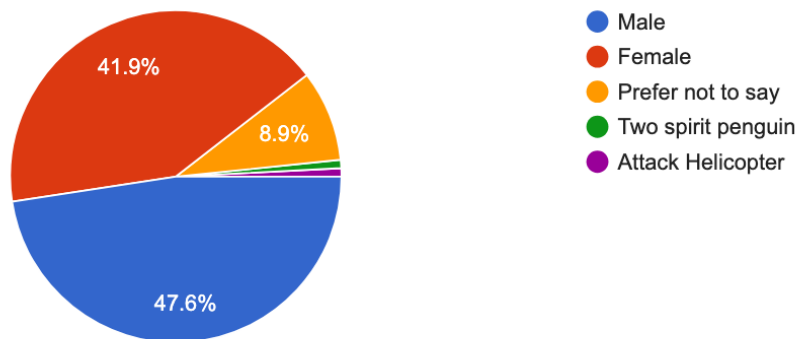


Figure 3.3: Gender Distribution of Respondents

Gender Distribution: Figure 3.3 shows a relatively balanced gender distribution with 47.6% male, 41.9% female, and 10.5% preferring not to disclose or other responses.

What best describes your current status?

124 responses

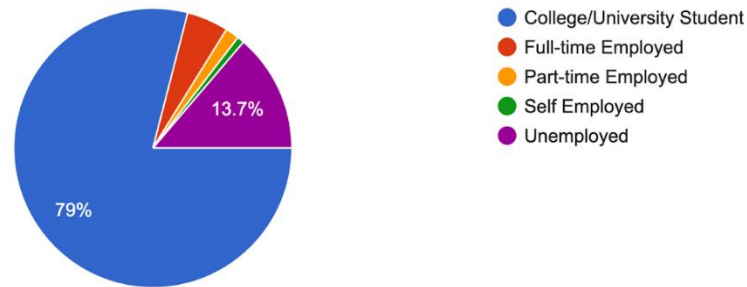


Figure 3.4: Status Distribution of Respondents

Status Distribution: As illustrated in Figure 3.4, the largest group was university students (79.0%), followed by unemployed individuals (13.7%), full-time employed (4.8%), part-time employed (1.6%), and self-employed (0.8%).

These demographic results confirm that the survey successfully reached the intended target audience — primarily university students aged 18-24.

3.3.2 Current Exercise Habits

How would you describe your current fitness level?

124 responses

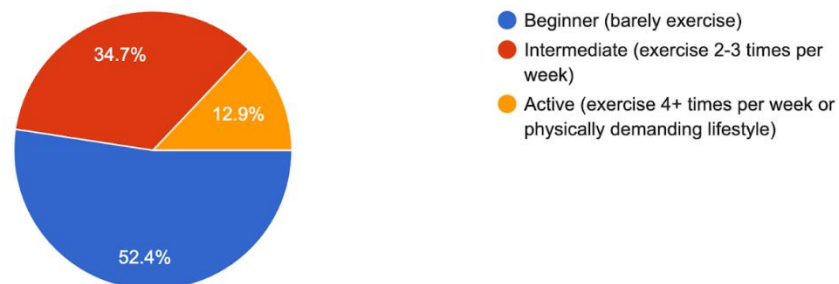


Figure 3.5: Fitness Level Distribution

Fitness Level: As shown in Figure 3.5, more than half of the respondents (52.4%) identified as beginners who barely exercise, while 34.7% were intermediate (exercising 2-3 times per week), and only 12.9% were active (4+ times per week).

How often do you currently exercise per week?

124 responses

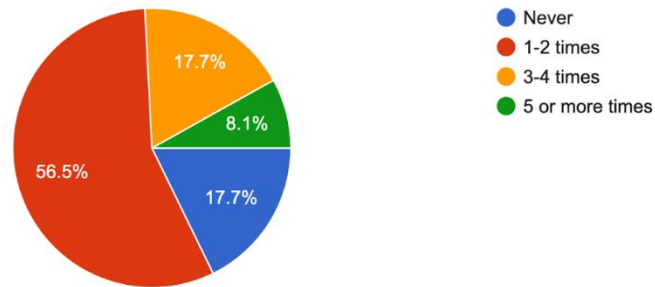


Figure 3.6: Exercise Frequency per Week

Exercise Frequency: Figure 3.6 illustrates that the majority (56.5%) exercise only 1-2 times per week, while 17.7% never exercise and another 17.7% exercise 3-4 times. Only 8.1% exercise 5 or more times per week.

Where do you usually exercise?

124 responses

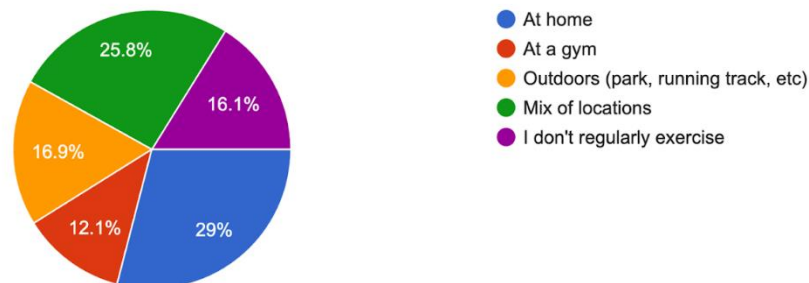


Figure 3.7: Exercise Location of Respondents

Exercise Location: As shown in Figure 3.7, the most common exercise locations were at home (29.0%) and a mix of locations (25.8%), followed by outdoors (16.9%). Only 12.1% exercise at a gym, indicating that the application must support home-based and equipment-free workouts.

These findings indicate that the majority of target users are beginners with infrequent exercise habits who prefer exercising at home or varied locations rather than gyms.

3.3.3 Fitness App Experience

Which fitness app(s) have you used? (Select all that apply)

124 responses

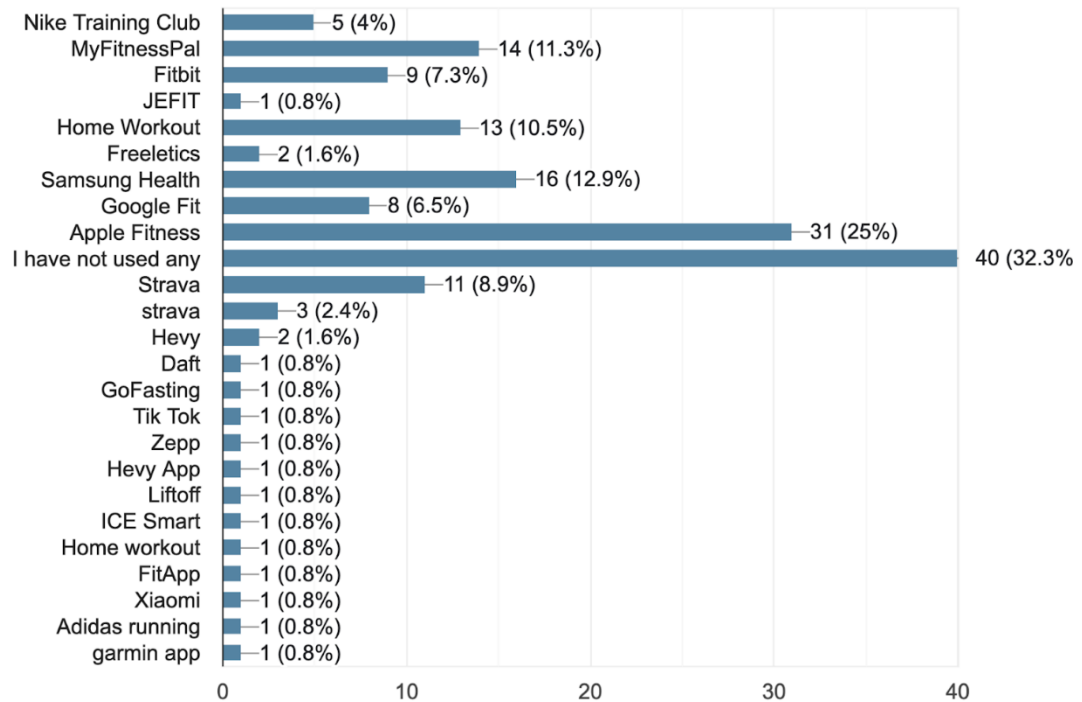


Figure 3.8: Fitness App Previously Used by Respondents

Previous App Usage: As shown in Figure 3.8, a significant portion of respondents (32.3%) had never used a fitness app, representing an opportunity for new user acquisition. Among those who had used apps, Apple Fitness (25.0%) was most popular, followed by Samsung Health (12.9%), MyFitnessPal (11.3%), and Home Workout (10.5%).

If you stopped using a fitness app, what was the main reason?

124 responses

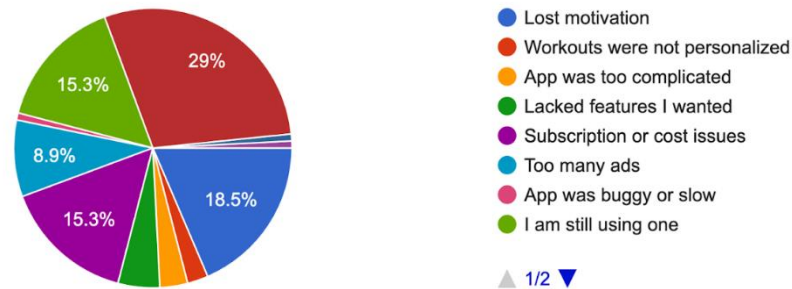


Figure 3.9: Reasons for Abandoning Fitness Apps

Reasons for Abandonment: Figure 3.9 shows the reasons respondents stopped using fitness apps. Excluding those who never used one (29.0%) or are still using (15.3%), the top reasons were: lost motivation (18.5%), subscription/cost issues (15.3%), and too many ads (8.9%). This highlights the importance of maintaining user motivation and offering a free, ad-minimal experience.

3.3.4 Barriers to Exercise

Respondents were asked to identify their top 3 barriers to regular exercise. As shown in Figure 3.10, the results strongly align with findings from the literature review (Brown et al., 2024; Mahmood et al., 2023):

What are your biggest barriers to exercising regularly? (Select up to 3)

124 responses

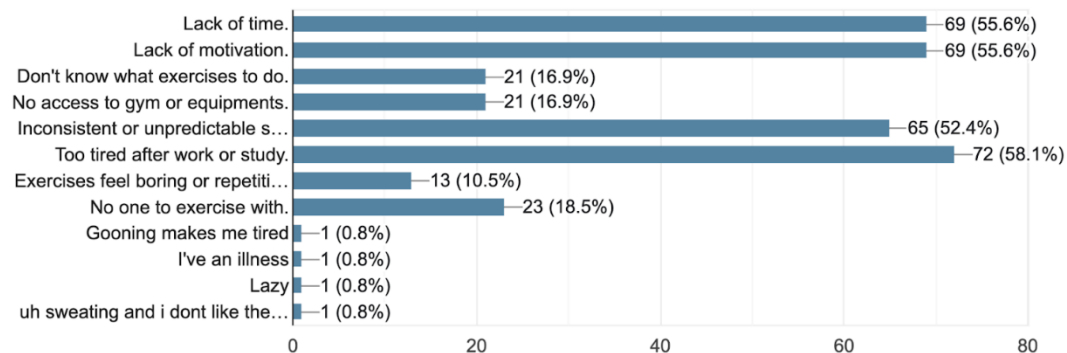


Figure 3.10: Barriers to Regular Exercise (Multiple Selection)

These findings align with barriers identified by Brown et al. (2024) and Mahmood et al. (2023), who found that time constraints, fatigue, and lack of motivation are primary deterrents to regular exercise among young adults.

If an app could solve ONE barrier for you, which would be most valuable?

124 responses

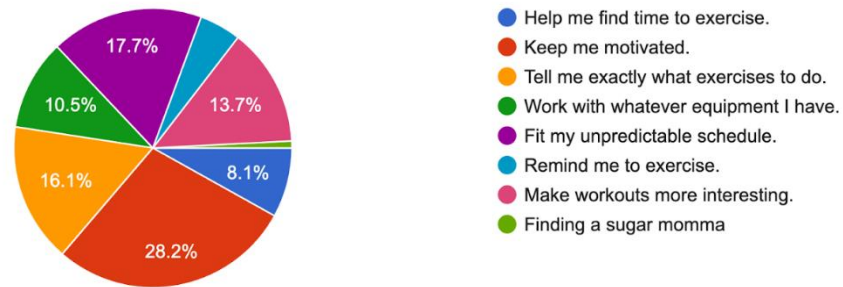


Figure 3.11: Most Valuable Solution an App Could Provide

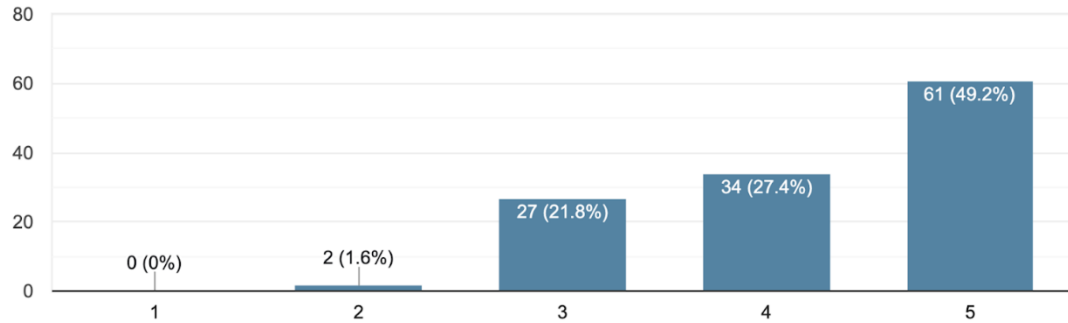
Most Valuable Solution: When asked which single barrier an app could most valuably solve (Figure 3.11), "Keep me motivated" was the top response (28.2%), followed by "Fit my unpredictable schedule" (17.7%) and "Tell me exactly what exercises to do" (16.1%). This validates the need for motivational features, flexible scheduling, and clear workout guidance in the application.

3.3.5 Feature Importance

Respondents rated the importance of 13 proposed features on a scale of 1 (Not Important) to 5 (Very Important). All features received mean scores above 4.0, indicating strong overall support. Figures 3.12 and Figure 3.13 show the results ranked by mean score.

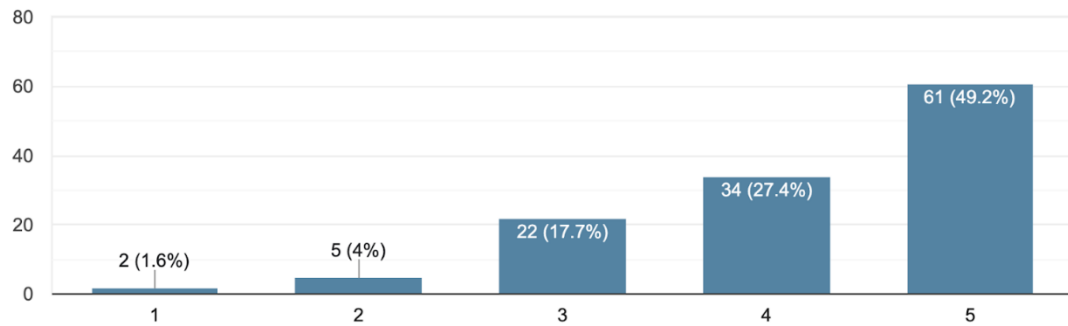
Personalized workout plans generated based on your fitness level, goals, available equipment, and time.

124 responses



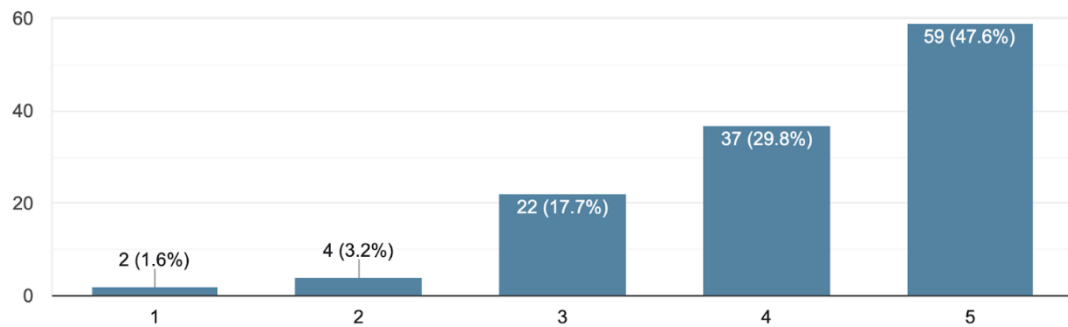
Ability to set your preferred workout days per week.

124 responses



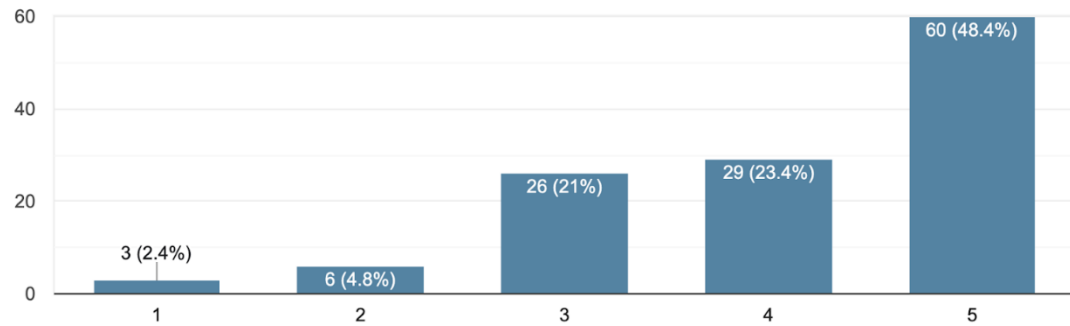
Ability to set your preferred time of day for workouts.

124 responses

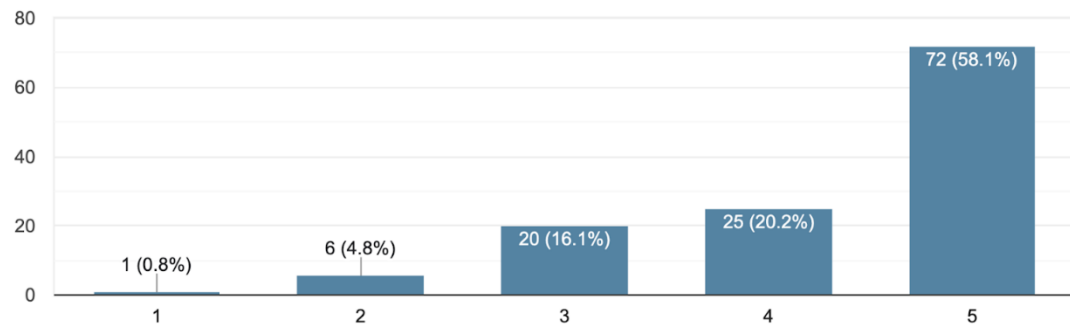


Workout reminders/notifications on your scheduled workout days.

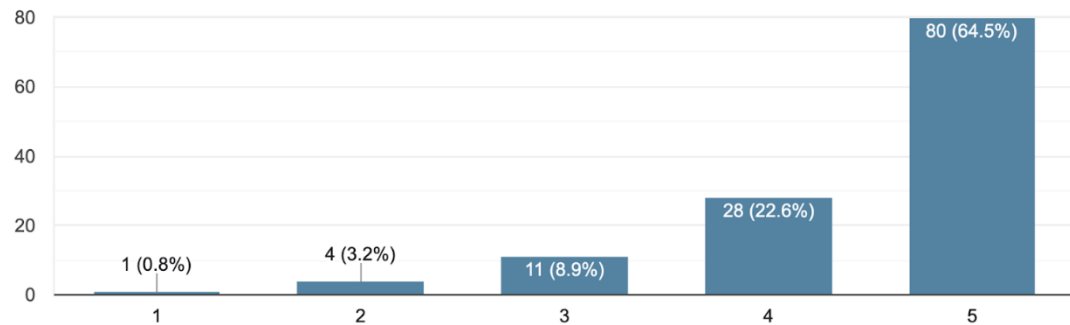
124 responses

**Calendar view to track and view your completed workouts.**

124 responses

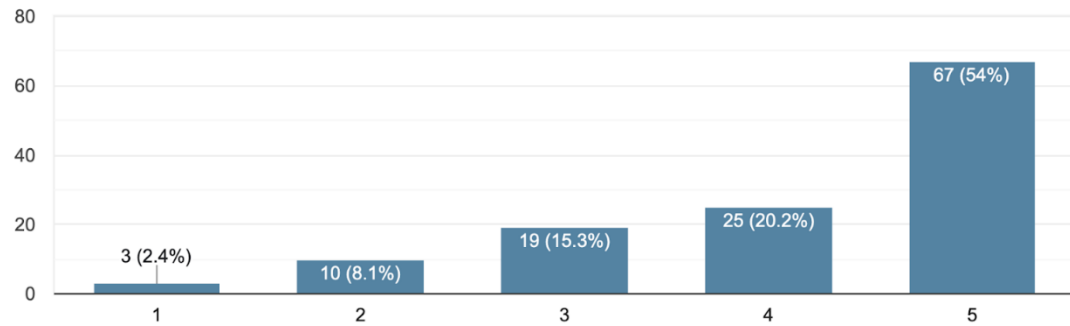
**Progress statistics (total workouts, current streak, weekly summary).**

124 responses

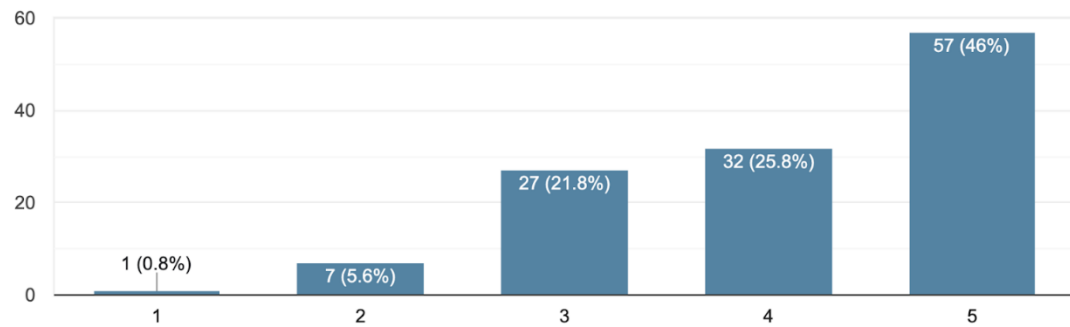


Achievement badges for milestones (e.g., first workout, 7-day streak, 30 workouts).

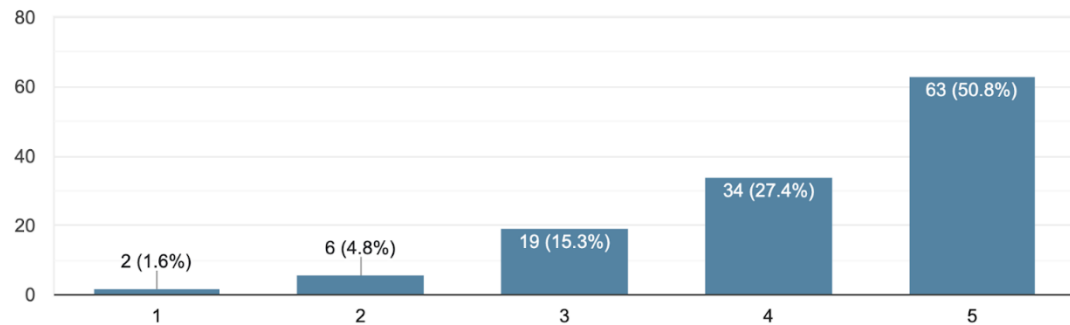
124 responses

**Exercise library with text instructions for each exercise.**

124 responses

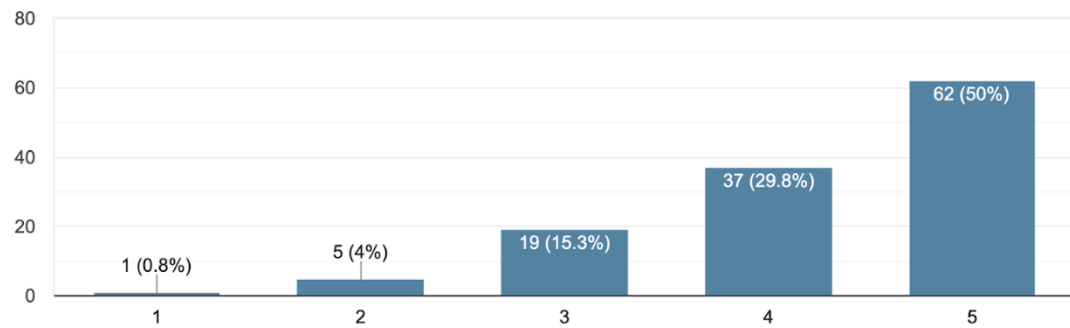
**Video tutorials (YouTube links) showing how to perform exercises.**

124 responses

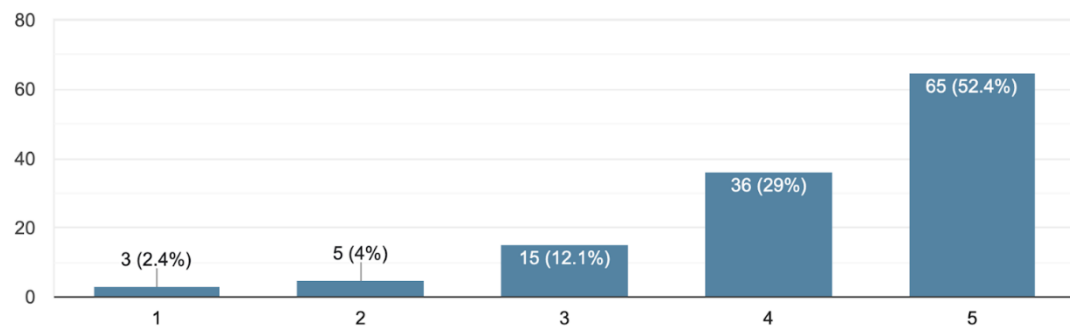


Ability to regenerate or get a new workout plan if not satisfied.

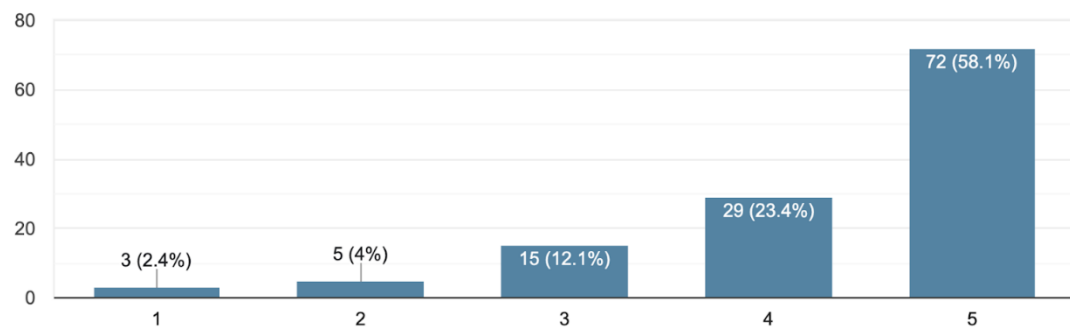
124 responses

**Ability to manually adjust or swap exercises in your plan.**

124 responses

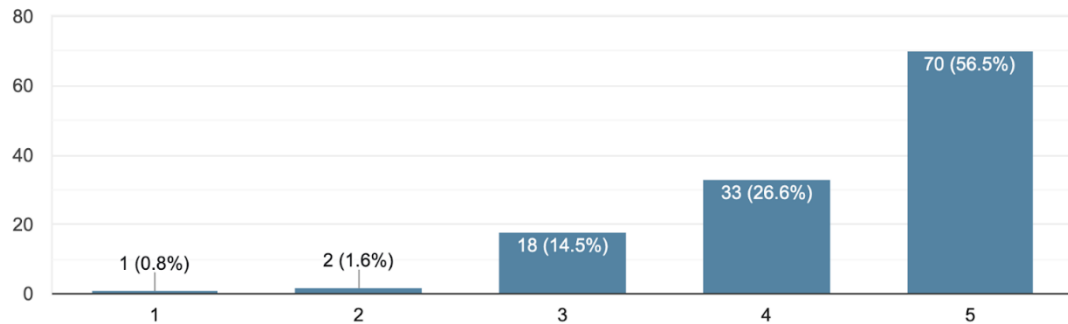
**Rest timer with alarm between sets.**

124 responses



Suggested starting weights based on your fitness level.

124 responses



Figures 3.12: Feature Importance Ratings - Mean Scores (1-5 Scale)

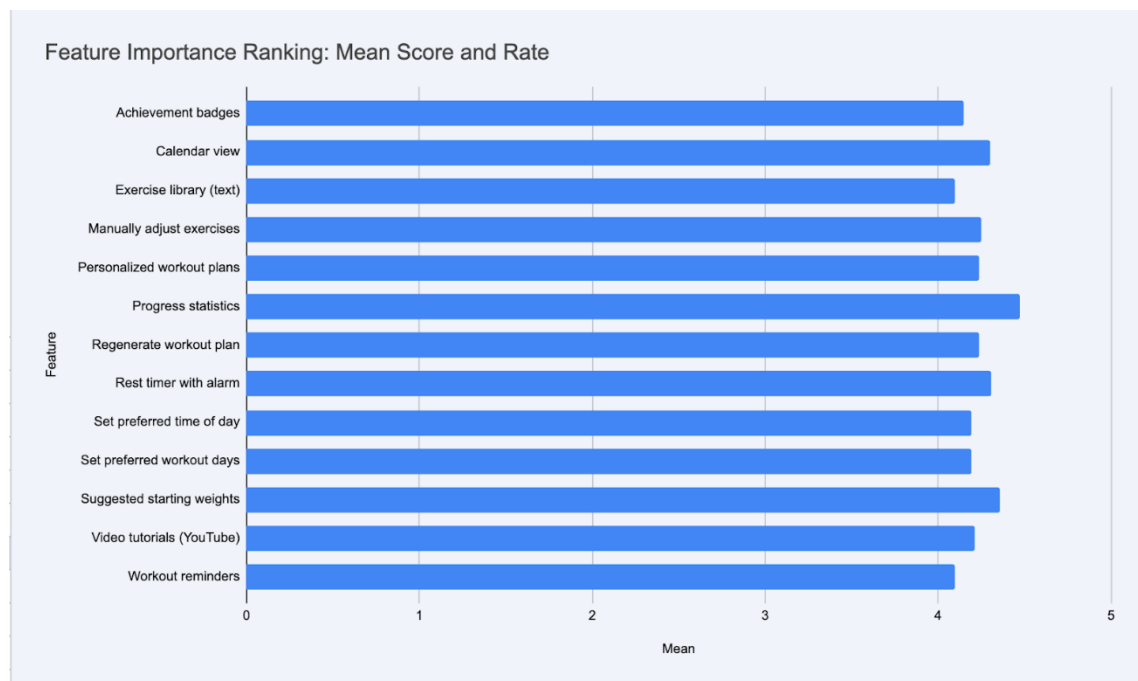


Figure 3.13: Horizontal Bar Chart for Feature Importance Ranking

The highest-rated feature was Progress Statistics (Mean: 4.47, 87.1% rated 4-5), confirming that users are highly motivated by tracking their progress. Notably, Rest Timer with Alarm (Mean: 4.31) and Suggested Starting Weights (Mean: 4.36) ranked highly, suggesting these features should be added to the requirements.

3.3.6 Workout Preferences

What is your primary fitness goal?

124 responses

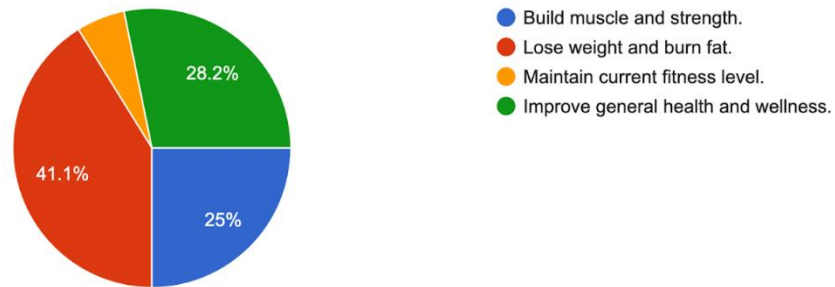


Figure 3.14: Primary Fitness Goal of Respondents

Primary Fitness Goal: As shown in Figure 3.14, the most common goal was "Lose weight and burn fat" (41.1%), followed by "Improve general health" (28.2%), "Build muscle and strength" (25.0%), and "Maintain fitness" (5.6%).

How many days per week would you ideally like to work out?

122 responses

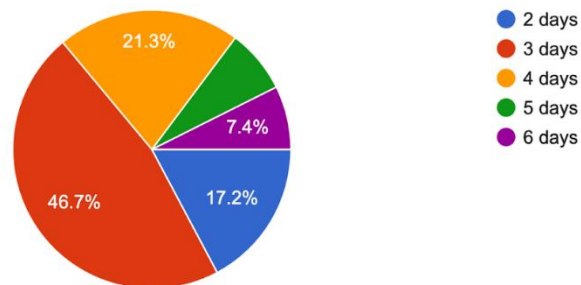


Figure 3.15: Preferred Workout Days per Week

Preferred Days per Week: Figure 3.15 shows that nearly half (46.0%) prefer to workout 3 days per week, followed by 4 days (21.0%) and 2 days (16.9%).

What is your preferred workout duration per session?

124 responses

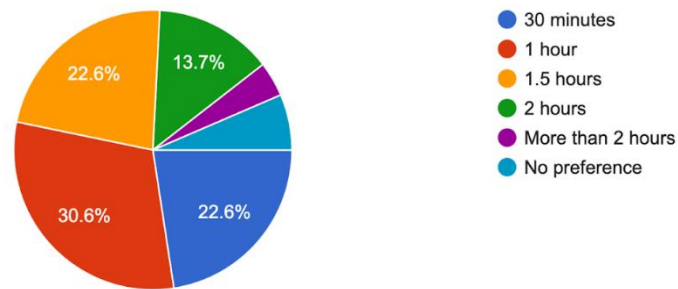


Figure 3.16: Preferred Workout Duration per Session

Preferred Duration: As illustrated in Figure 3.16, the most preferred durations were 1 hour (30.6%), 1.5 hours (22.6%), and 30 minutes (22.6%), indicating users want time-efficient workouts.

What equipment do you have access to? (Select all that apply)

124 responses

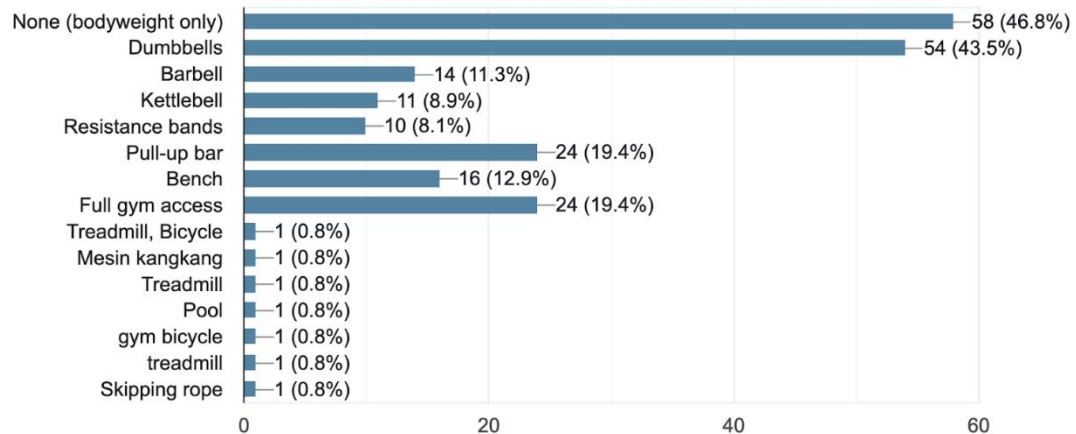


Figure 3.17: Equipment Access (Multiple Selection)

Equipment Access: Figure 3.17 shows that a significant portion (46.8%) have access to bodyweight only, while 43.5% have dumbbells. Only 19.4% have full gym access. This confirms the need for equipment-flexible workout generation.

What time of day do you prefer to work out? (Select all that apply)

124 responses

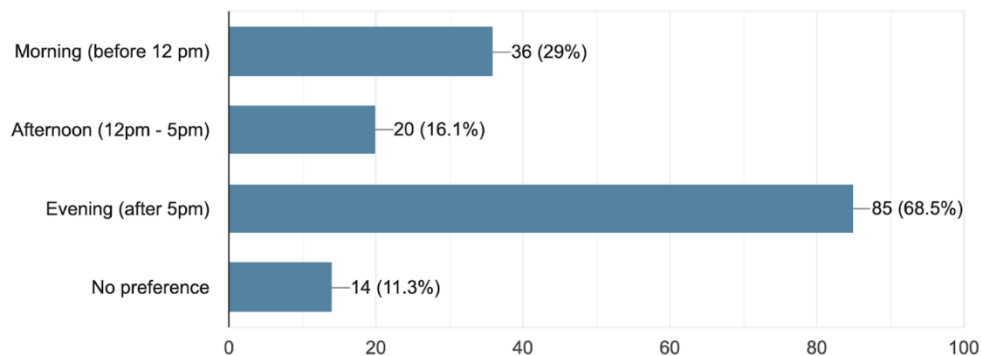


Figure 3.18: Preferred Time of Day for Workouts

Preferred Time of Day: As shown in Figure 3.18, the majority (68.5%) prefer evening workouts (after 5pm), with morning (29.0%) as a secondary preference.

3.3.7 App Preferences

If the app generates a workout plan you're not satisfied with, what would you prefer?

124 responses

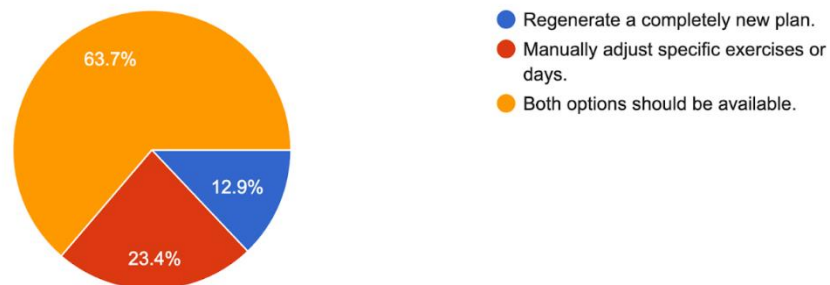


Figure 3.19: Preferred Method When Unsatisfied with Plan

Plan Regeneration: When asked about handling plan dissatisfaction (Figure 3.19), 63.7% wanted both options available (regenerate new plan AND manually adjust), 23.4% preferred manual adjustment only, and 12.9% preferred complete regeneration.

Would you find it helpful if the app suggested starting weights based on your fitness level and past history?

124 responses

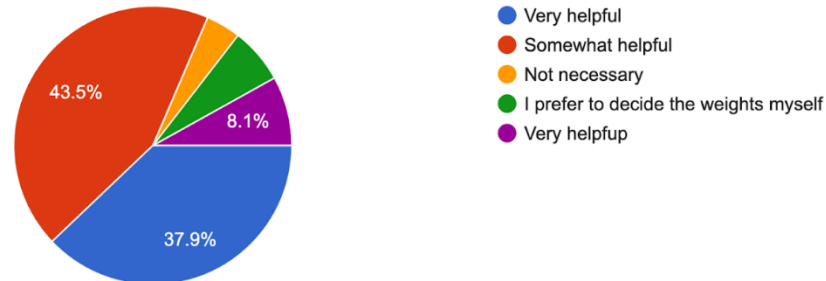


Figure 3.20: Helpfulness of Suggested Starting Weights

Weight Suggestions: As shown in Figure 3.20, 89.5% of respondents found suggested starting weights helpful (43.5% somewhat helpful, 46.0% very helpful), validating this feature.

How do you feel about workout reminders/notifications?

124 responses

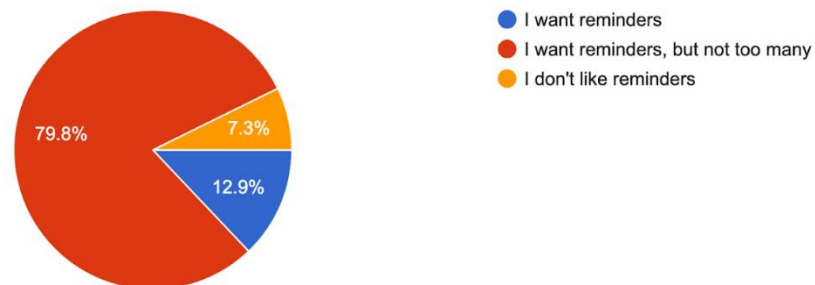


Figure 3.21: Workout Reminder Preferences

Reminders: Figure 3.21 shows that the majority (79.8%) want reminders but "not too many," while 12.9% want regular reminders and only 7.3% dislike reminders entirely. This supports implementing non-intrusive reminders on workout days only.

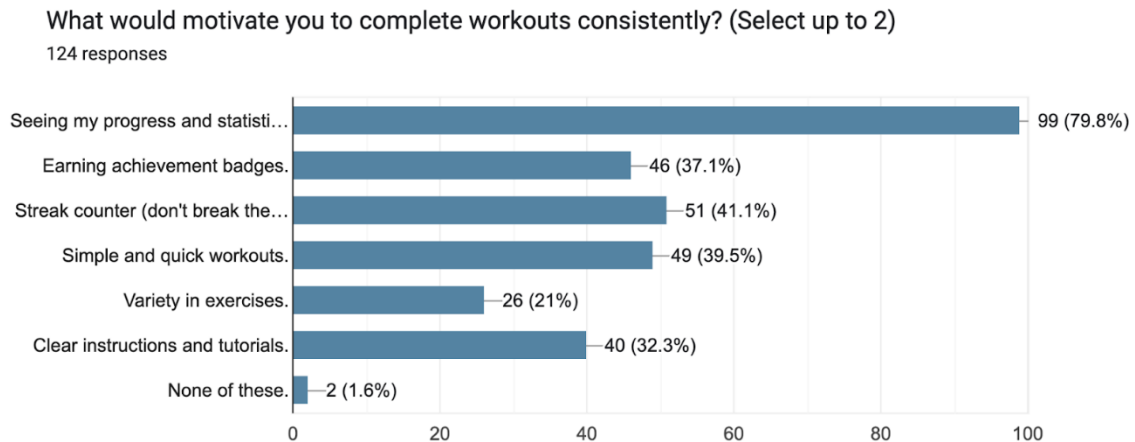


Figure 3.22: Factors That Would Motivate Consistent Workouts

Motivation Factors: When asked what would motivate consistent workouts (Figure 3.22), the top factors were: seeing progress and statistics (79.8%), streak counter (41.1%), simple and quick workouts (39.5%), earning achievement badges (37.1%), and clear instructions (32.3%).

3.3.8 Open Feedback

Respondents provided additional suggestions through open-ended questions. Key themes included:

Features to help stick with app long-term:

- Simple, easy-to-use interface
- Minimal or no advertisements
- Personalized workouts
- Video tutorials for exercises
- Motivational messages and encouragement

Suggested additional features:

- Diet and meal planning integration
- Specific body part targeting options
- Built-in music integration
- Fitness journal or diary
- Daily motivational quotes

These suggestions will be considered for future enhancements beyond the current project scope.

3.4 Requirements Specification

Based on the literature review findings (Chapter 2) and survey analysis (Section 3.3), the following requirements have been defined for the Smart Workout Planner & Tracker mobile application. Requirements are prioritized as Must Have, Should Have, or Nice to Have based on survey support and core functionality.

3.4.1 Functional Requirements

Functional requirements define the specific features and functions the system must perform. Table 3.2 presents the functional requirements derived from survey findings.

ID	REQUIREMENT	PRIORITY	SURVEY SUPPORT
FR01	Users of the system will receive personalized fitness plans based on their fitness levels, goals, access to equipment, frequency of workouts, and/or desired duration of workouts..	Must Have	76.6% rated 4-5
FR02	Users of the system will be able to specify their preferred days of the week and times of day to work out, as well as how long they want their workouts to last.	Must Have	76.6% rated 4-5
FR03	Users of the system will receive reminders on the days they are scheduled to work out.	Should Have	71.8% rated 4-5; 92.7% want reminders
FR04	Users of the system will have access to a calendar to record the workouts that they did complete.	Must Have	78.2% rated 4-5
FR05	Achievement badges will be displayed by the system for milestones such as the first completed workout and for working out for seven consecutive days.	Should Have	74.2% rated 4-5; 37.1% motivated by badges
FR06	Users of the system will be able to see statistics showing their total number of completed workouts, their current streak of workouts, and a summary of their weekly workouts.	Must Have	87.1% rated 4-5 (Highest rated)
FR07	An exercise library that includes text-based instructions for each exercise, as well as links to corresponding YouTube videos, will be provided by the system.	Must Have	78.2% rated 4-5

FR08	Users of the system will be able to create a user account and securely log into that account.	Must Have	Core functionality
FR09	Users of the system will be able to edit their profiles and preferences.	Must Have	Core functionality
FR10	There will be a timer function provided by the system to allow the users rest time before continuing with their workout.	Should Have	81.5% rated 4-5
FR11	Users of the system will be able to make manual changes to or alternate exercises in their fitness plans.	Should Have	81.5% rated 4-5; 63.7% want this option
FR12	Users of the system will have the ability to have a newly generated plan of exercises created.	Should Have	79.8% rated 4-5
FR13	The system will suggest exercise starting weights for users of the system based on their level of fitness.	Nice to Have	83.1% rated 4-5; 89.5% find helpful
FR14	After completing a workout, users of the system will be able to rate how difficult they thought it was.	Nice to Have	71.8% want to rate

Table 3.2: Functional Requirements Derived From Survey Responses

3.4.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system. Table 3.3 presents the non-functional requirements.

ID	REQUIREMENT	CATEGORY
NFR01	The system shall load workout plans within 3 seconds.	Performance
NFR02	The system shall be available 99% of the time.	Availability
NFR03	The system shall encrypt user data in transit and at rest.	Security
NFR04	The system shall be compatible with Android devices (API level 24+).	Compatibility
NFR05	The user interface shall be intuitive and require minimal training.	Usability
NFR06	The system shall handle at least 100 concurrent users.	Scalability

NFR07	The system shall follow modular architecture for maintainability.	Maintainability
-------	---	-----------------

Table 3.3: Non-Functional Requirements Derived From Survey Responses

3.4.3 User Requirements

User requirements describe what users need to accomplish using the system. Table 3.4 presents the user requirements derived from survey responses.

ID	Requirement	Survey Evidence
UR01	Users will be allowed to create fitness objectives (muscle gain, fat loss, maintain weight, or stay healthy).	Q17: 41.1% lose weight, 28.2% health, 25.0% build muscle
UR02	Users can identify what gym equipment they have access to.	Q20: 46.8% bodyweight only, 43.5% dumbbells
UR03	Users can specify which days they want to work out over the course of a week.	Q18: 46.0% prefer 3 days
UR04	Users should indicate their preferred length of time they would like to exercise on a daily basis.	Q19: 30.6% prefer 1 hour, 22.6% prefer 30 mins
UR05	Users will be able to see when they complete workouts.	Core functionality
UR06	Users can earn badges and get feedback for consistency in using the app.	Q22: 37.1% motivated by badges
UR07	Users will be able to see their previous workouts via a calendar format.	Q feature: 78.2% rated 4-5
UR08	Users will be able to view exercise instructions with instructional videos.	Q feature: 78.2% rated 4-5
UR09	Users will be permitted to use a timer to rest between sets.	Q feature: 81.5% rated 4-5

UR10	Users will have the ability to change/trade out/modify gen plans of action.	Q23: 63.7% want both options
------	---	------------------------------

Table 3.4: User Requirements Derived From Survey Responses

3.5 Summary

This chapter presented the requirements analysis for the Smart Workout Planner & Tracker mobile application. An online survey was conducted using Google Forms and distributed via social media platforms, collecting 124 valid responses primarily from university students aged 18-24.

Key Findings:

Aspect	Finding
Target Audience	90.3% aged 18-24; 79.0% university students — target validated
Fitness Level	52.4% beginners; 56.5% exercise only 1-2 times per week
Exercise Location	29.0% at home; only 12.1% at gym — supports equipment-flexible workouts
Top Barriers	Too tired (58.1%), lack of time (55.6%), lack of motivation (55.6%)
Most Wanted Solution	Keep me motivated (28.2%), fit my schedule (17.7%), tell me what to do (16.1%)
Top Feature	Progress statistics (87.1% rated 4-5)
Workout Preference	3 days/week (46.0%), 1 hour sessions (30.6%), evening time (68.5%)
Equipment Reality	46.8% bodyweight only — must support equipment-free workouts
Reminders	92.7% want reminders, but "not too many" (79.8%)

Top Motivator	Seeing progress and statistics (79.8%)
----------------------	--

Table 3.5: Key Findings From Survey

The findings represented by the 14 functional requirements, 7 non-functional requirements and 10 user requirements were then prioritised as Must Have with percentages of support from the survey and how many of these were considered core functionality to produce a list of these items as Should Have and Nice To Have.

Following this chapter is the system design (consisting of system architecture, use case diagrams, class diagrams and activity diagrams) based on the identified requirements.

Chapter 4: System Design

4.1 Overview

The Smart Workout Planner & Tracker application converts the functional and non-functional requirements documented in Chapter 3 into a design of the system. The design, therefore, will establish both the structure and behaviour of the application and will provide the foundation for the implementation phase of the project.

4.2 Architecture Diagram

The Architecture Diagram of the Smart Workout Planner & Tracker application provides an overview of the app's entire architecture, which consists of three tiers: front-end tier, back-end tier and externally provided services-tier.

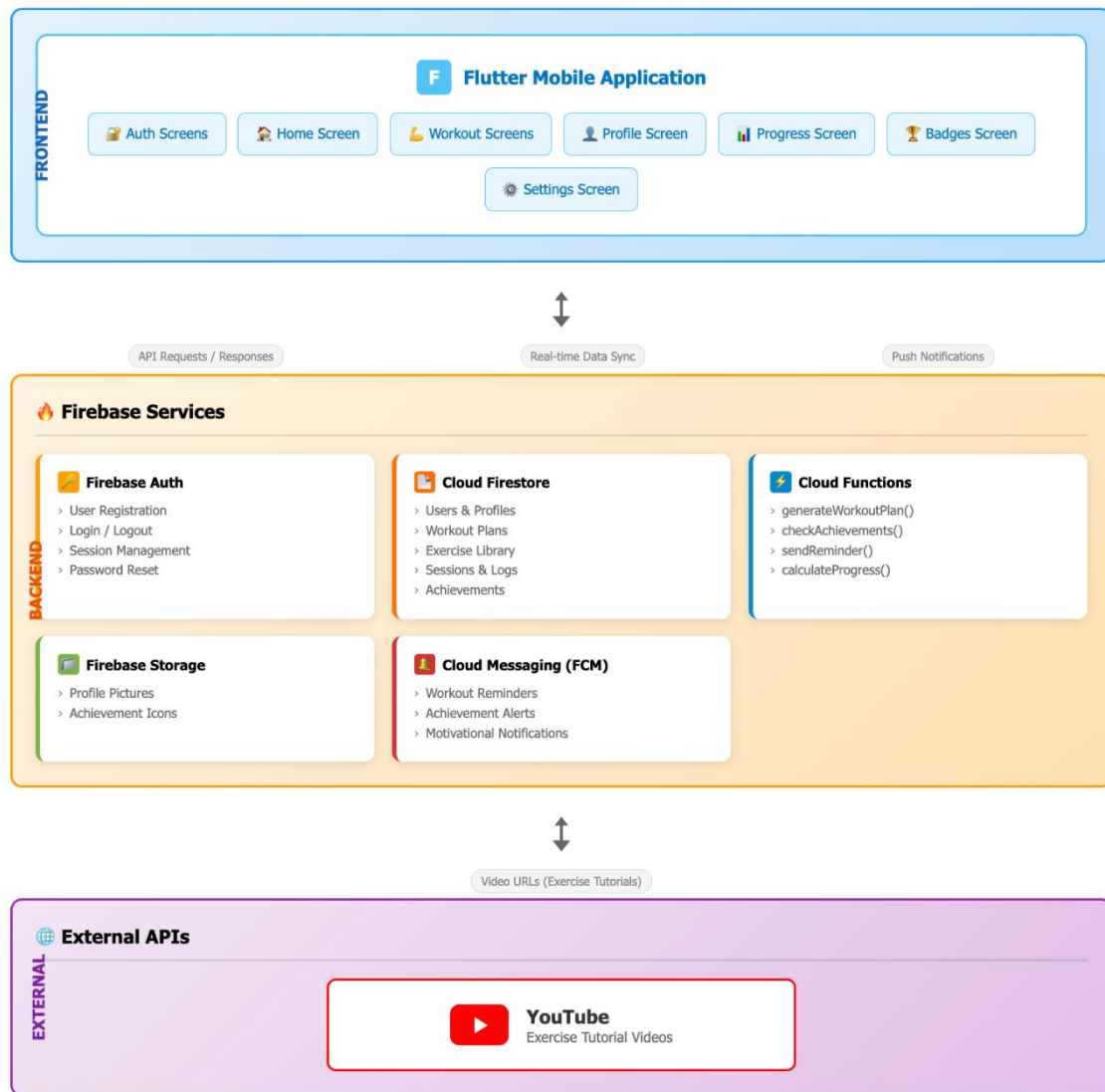


Figure 4.1: Architecture Diagram

Figure 4.1 presents the system architecture organized into three distinct layers:

1. Frontend Layer

Component	Technology	Description
-----------	------------	-------------

Mobile Application	Flutter	Provides user interface screens for the SWPT application. User interface screens include Authentication, Home, Workout, Profile, Progress, Badges and Settings.
---------------------------	---------	---

Table 4.1: Frontend Layer of Architecture Diagram

2. Backend Layer

Service	Purpose
Firestore Auth	Enables user registration, login, logout, session management and user password reset functionality.
Cloud Firestore	Provides a NoSQL data store which is where all SWPT application data will be stored. All SWPT application data will include user data, profile data, workout plan data, exercise library data, session logs and achievement data.
Cloud Functions	Provides a serverless execution environment for business logic including the workout plan generation algorithm, achievement verification and scheduled reminder sending.
Firestore Storage	Object storage where a user can upload content that is related to an achievement such as a profile photo and/or an achievement badge icon.
Firestore Cloud Messaging (FCM)	Push notification service that sends workout reminders, achievement notifications and motivational messages to the user.

Table 4.2: Backend Layer of Architecture Diagram

3. External API's Layer

Service	Purpose
YouTube	YouTube hosts exercise tutorial videos that are embedded into each exercise's instruction.

Table 4.3: External API's Layer of Architecture Diagram

The architecture has been designed based on a 'Serverless' model using Firebase managed services to reduce the burden of maintaining physical infrastructure, while providing future scalability. The Flutter app communicates with Firebase services via real-time data sync (Firestore), RESTful API calls (Cloud Functions) and push notifications (FCM) giving the app's users offline capabilities using Firestore's local caching methods so that they can view their planned workouts without the need for internet access.

There is a bi-directional data flow between both the Front-End and Back-End layers, with the Cloud Functions performing heavy lifting by generating and checking the user's workout plan and achievements so that the compute load has been removed from the mobile device.

4.3 Use Case Diagram

The use case diagram shows the functional requirements of the Smart Workout Planner & Tracker System and gives identification of the primary actor along with the many interactions that can be made by the primary actor in the system organized by six functional categories.

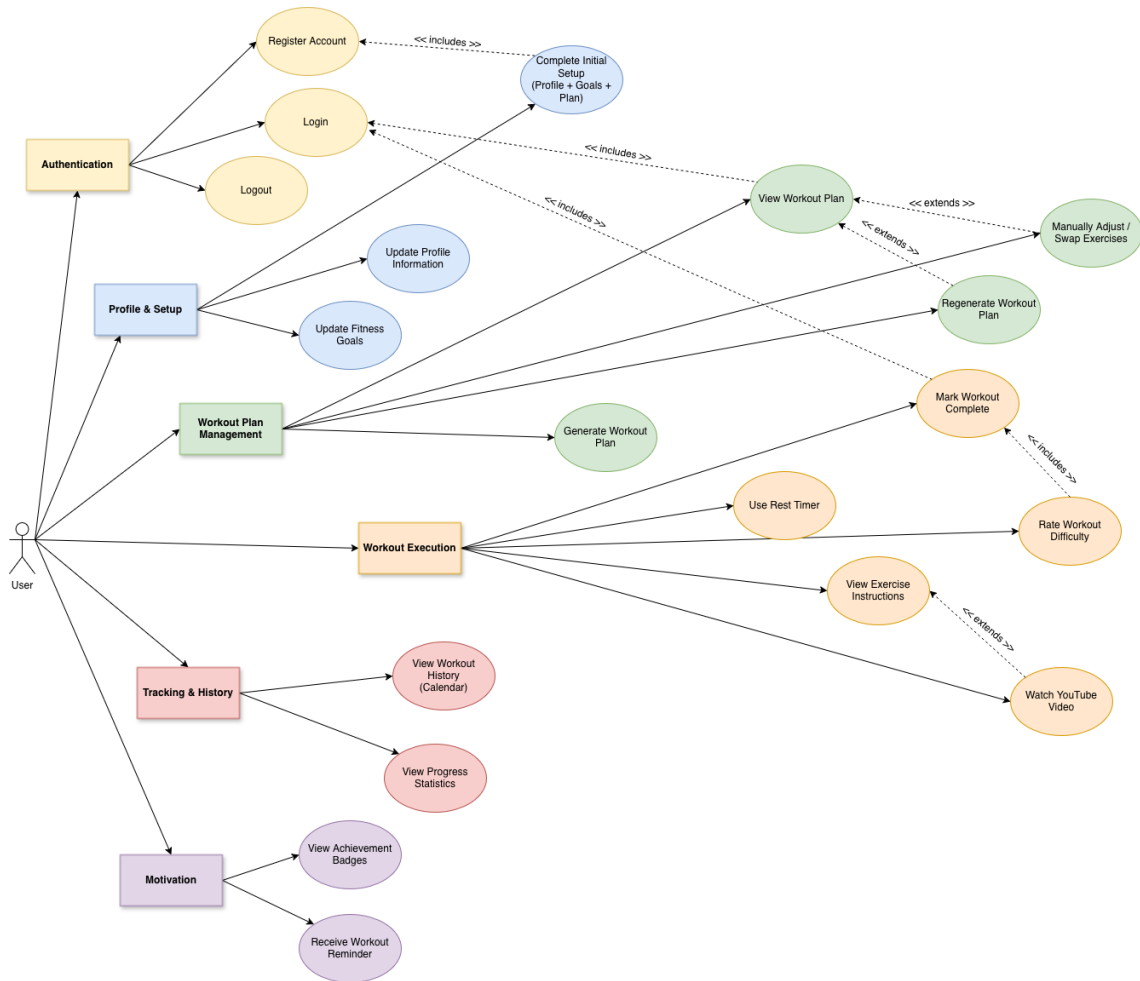


Figure 4.2: Use Case Diagram

Figure 4.2 presents the use case diagram for the system, showing a single primary actor (User) interacting with 18 use cases. The use cases are grouped into the following functional categories:

Category	Use Cases	Description
Authentication	Register Account, Login, Logout	Handles user account creation and session management

Profile & Setup	Complete Initial Setup, Update Profile Information, Update Fitness Goals	Manages user onboarding and preference configuration
Workout Plan Management	Generate Workout Plan, View Workout Plan, Manually Adjust/Swap Exercises, Regenerate Workout Plan	Enables creation, viewing, and modification of personalized workout plans
Workout Execution	View Exercise Instructions, Watch YouTube Video, Use Rest Timer, Mark Workout Complete, Rate Workout Difficulty	Supports the user during active workout sessions
Tracking & History	View Workout History (Calendar), View Progress Statistics	Provides historical data and progress visualization
Motivation	View Achievement Badges, Receive Workout Reminder	Implements gamification and reminder features to encourage adherence

Table 4.3: Use Case Descriptions

The diagram also illustrates the relationships between use cases using UML stereotypes:

- **Include relationships:**

The use case diagram also shows the relationships between use cases using UML Stereotypes: include relationships indicate a mandatory relationship. For example, in order to complete the “Complete Initial Setup” use case, a user must register an account to complete their profile. This is shown in the “View Workout Plan” is dependent on the “Login” use case since there must be an authenticated user to access the workout plan. Another example of the use of include relationships is in the “Mark Workout Complete” use case, which is dependent on the “Rate Workout Difficulty” use case, because the system requests users to provide feedback on their workout at the end of each session.

- **Extend relationships:**

The extend relationships denote an optional relationship. The ability to “Manually Adjust/Swap Exercises” and to “Regenerate Workout Plan” extend from the “View Workout Plan” use case; therefore, users can choose to modify

their plan. Additionally, the “Watch YouTube Video” extends from the “View Exercise Instructions” use case and provides additional video support that users may wish to access as part of their exercise instruction

4.4 Activity Diagram

Activity diagrams depict how a system behaves over time by capturing the activities, including activity flows and decisions associated with certain key processes of that business area. Two activity diagrams were created to represent the two primary workflows of Smart Workout Planner & Tracker: User Onboarding and Generate Workout Plan.

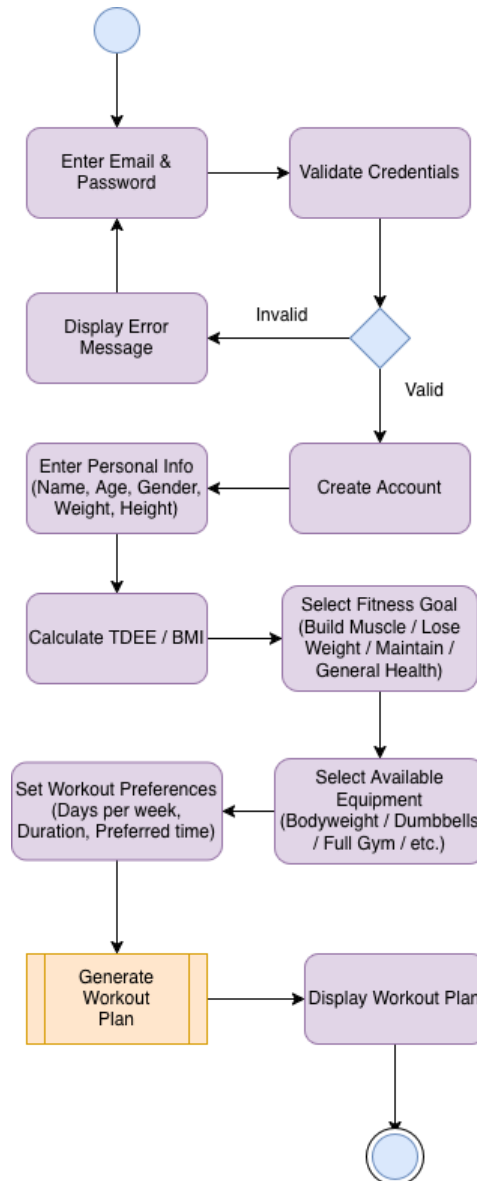


Figure 4.4: Activity Diagram - User Onboarding

Figure 4.3 shows the user onboarding diagram that starts at the point of sign-up for a user account. The process flow of that user onboarding activity is as follows:

1. A new user, per registration, creates an account by entering their email and password. The system verifies, or does not verify their registration credentials
2. If it does not verify, then the user is notified that the email and/or password is invalid and is requested to re-enter their registration information.

3. A user will be able to access their account if the system successfully verifies their registration, creating a new user account and getting the required user-to-personal The system uses this personal information to calculate the Total Daily Energy Expenditure (TDEE) and Body Mass Index (BMI) values for the user based upon that personal information that the user has entered into the system.

4. The user will select personal fitness goals (e.g., commitment level to working out), available equipment, and workout preferences (i.e., number of days per week that they wish to work out, the desired length of time that they want to work out, the preferred day of workout).

5. The system then generates a unique, personalized workout plan for the user that will then be displayed to them. As shown in the activity diagram, the User Onboarding activity diagram (user account creation) is calling the Generate Workout Plan subprocess (which is further illustrated in the diagram that follows).

The onboarding process calls the "Generate Workout Plan" subprocess, which is detailed in the following diagram.

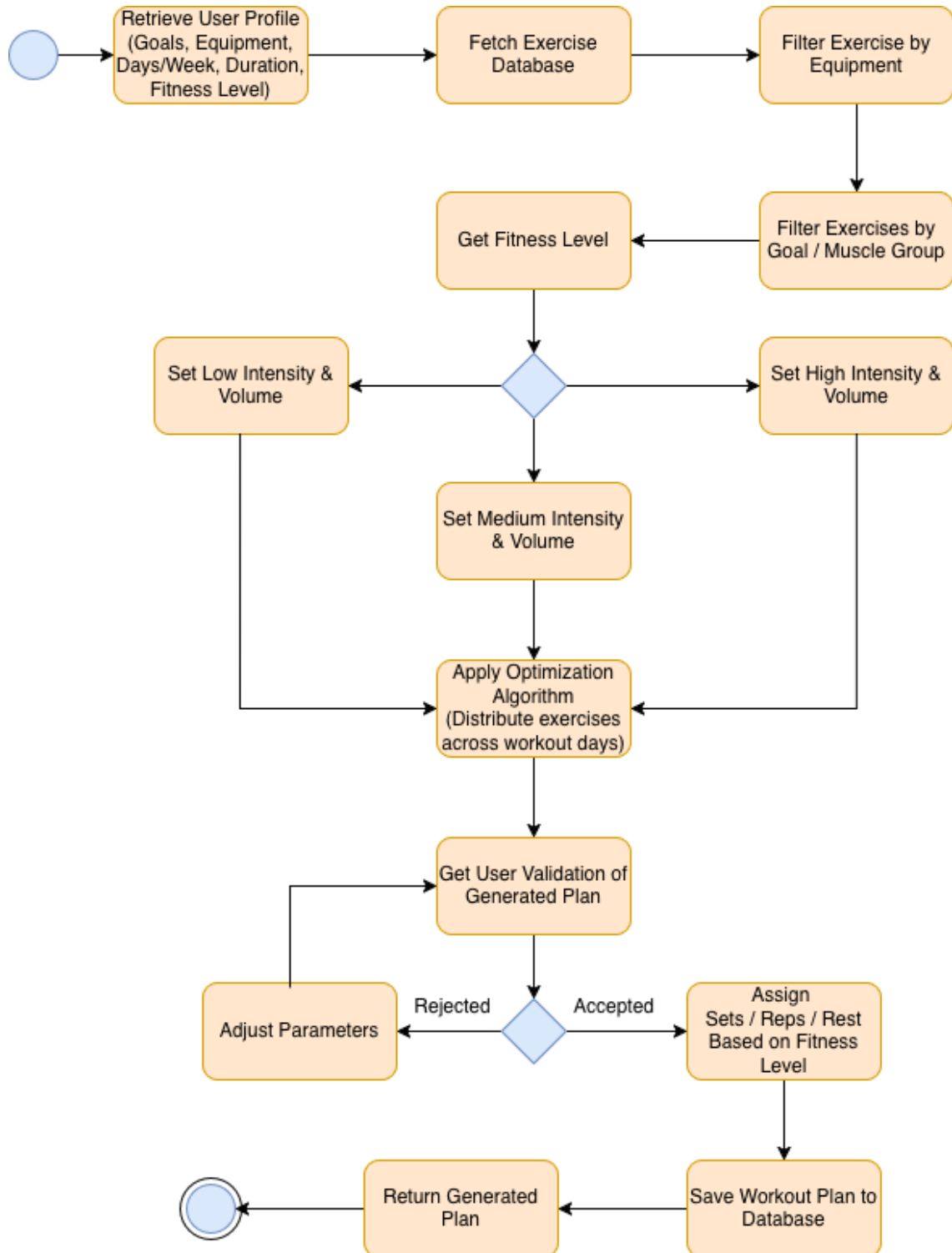


Figure 4.4: Activity Diagram - Generate Workout Plan

The diagram in Figure 4.4 depicts an expansion of the process through which a workout plan is generated. It includes the algorithmic logic of the personalization engine and is outlined below:

Phase	Activities	Description
Data Retrieval	Retrieve User Profile, Fetch Exercise Database	This step gathers user preferences as well as the available exercises.
Filtering	Filter by Equipment, Filter by Goal/Muscle Group	This step reduces the number of exercises based upon the constraints provided by the user.
Intensity Configuration	Get Fitness Level → Set Low/Medium/High Intensity & Volume	This step configures the workout parameters according to the user's fitness level.
Optimization	Apply Optimization Algorithm	This step uses an algorithmic process to evenly distribute the exercises across workout days.
Validation	Get User Validation → Accepted/Rejected	Users will have the opportunity to validate the generated workout plan. If the user does not accept the validation, the system will re-configure the parameters and re-optimize the plan, as necessary, until an acceptable workout plan is created.
Persistence	Assign Sets/Reps/Rest, Save to Database, Return Plan	This will finalize and store the user's generated workout plan.

Table 4.5: Phases of Workout Plan Generation Process

The validation process ensures that a user can reject their generated workout plan should it fall short of their expectations, allowing the system to adjust the parameters and regenerate until an acceptable workout plan is produced.

All of the activity diagrams within the invoiced documents illustrate the complete flow from registering a user to providing that user with a personalized, algorithmically-optimized workout plan developed for their specific goals, equipment, and individual fitness level.

Figure 4.5 presents the class diagram consisting of 15 entity classes organized into four functional groups

Category	Classes	Purpose/Description
User Management	User, Profile, FitnessGoal, WorkoutPreference, UserEquipment, Equipment	Used to maintain Users with their personal information, fitness goals, and equipment available.
Workout Planning	WorkoutPlan, WorkoutDay, WorkoutExercise, Exercise	Organises the created Workout Plans and Exercise Library.
Workout Tracking	WorkoutSession, ExerciseLog	Keeps track of Workout Completes and individual exercise performance.
Motivation	Achievement, UserAchievement, Reminder	Covers gamification badges and workout reminders

:

Table 4.6: Description of Classes From their Categories

The Class Diagram uses two forms of ownership to explain how Classes rely on each other:

- **Composition (black diamond):**

Indicates Strong Ownership, where Child Entity cannot exist without the Parent Entity. For example: A WorkoutDay is contained within a WorkoutPlan; if a WorkoutPlan is deleted, all WorkoutDays belonging to that WorkoutPlan will also be deleted. Likewise, an ExerciseLog will also be deleted if the Parent WorkoutSession it belongs to is deleted.

- **Aggregation (white diamond):**

Indicates weak ownership of child entities by their parent entities such that the child entities are able to exist independently from their parent entities. For example, the WorkoutExercise entity references the Exercise entity within the exercise library's exercise entities. However, if the WorkoutExercise entity is deleted, the Exercise entity can still exist within the exercise library. The

relationship between the Exercise entity and the Equipment entity also illustrates this example.

The User entity acts as the primary entity and will maintain relationships with Profile (1 to 1), FitnessGoal (1 to 1), WorkoutPreference (1 to 1), WorkoutPlan (1 to many), WorkoutSession (1 to many), Reminder (1 to many) and UserAchievement (1 to many) entities. The relationship between User and Equipment is many to many. The relationship is resolved by creating a UserEquipment junction table that allows the User(s) entity to create multiple UserEquipments as they relate to the provided available equipment.

There are defined methods (functions) in those User, Profile and WorkoutPlan entities which provide business activity (method) logic. Password verify is an example of the business activity (method) within the User entity, while providing means for the Profile entity to update information is an example of business activity (method) within the Profile entity; and as in prior examples, the WorkoutPlan entity provides methods for activating and archiving workout plans. All other entities primarily serve the role of the data entities and map their attributes to database columns.

4.6 Entity Relationship Diagram

Database schema for Smart Workout Planner & Tracker system defined by the Entity Relationship Diagram (ERD). The ERD defines the tables, attributes, primary keys, foreign keys, and relationships required for data to be persisted in the system.

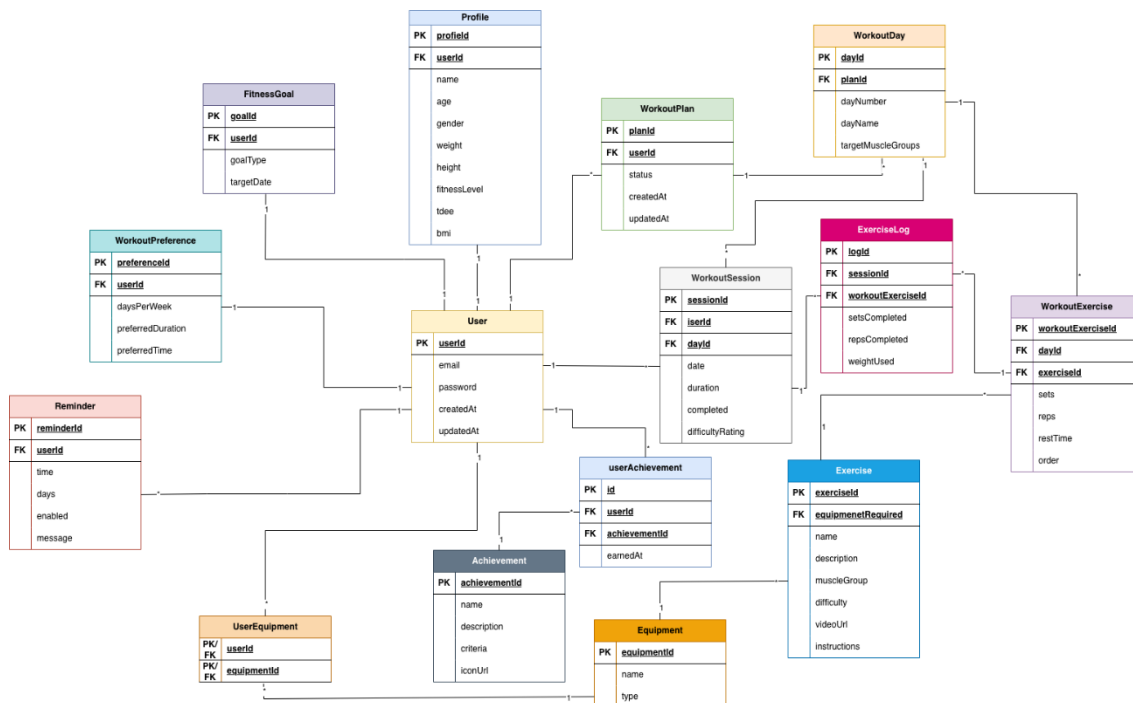


Figure 4.6: Entity Relationship Diagram

Figure 4.6 presents the ERD consisting of 15 entities organized into five logical groups:

Category	Entities	Purpose/Description
User Management	User, Profile, FitnessGoal, WorkoutPreference, UserEquipment, Reminder	Stores user account, personal information, and preferences.
Workout Planning	WorkoutPlan, WorkoutDay, WorkoutExercise	Structures the generated personalized workout plans
Workout Tracking	WorkoutSession, ExerciseLog	Completed workouts will be recorded and performance will be tracked.
Motivation	Achievement, UserAchievement	Manage progress of users through rewards and badges that encourage them to work out.
Library	Equipment, Exercise	Holds all data related to exercises and available equipment.

Table 4.7: Description of Entities From Their Categories

Cardinalities of Relationships in the ERD are defined as follows:

- **One-to-One (1:1):** Each individual user will have a single relationship with a Profile, Fitness Goal, and Workout Preference; thus, every user will have only one set of data regarding these factors.
- **One-to-Many (1:*):** A user can have multiple Workout Plans, Workout Sessions, Reminders, and User Achievements. There will be multiple Workout Days associated with each Workout Plan, and multiple Workout Exercises will be associated with each Workout Day.

- **Many-to-One (*:1):** There will be multiple Workout Exercises associated with the same Exercise within the Library, and there will be multiple Exercises that can be performed with the same piece of Equipment.
- **Many-to-Many (*:*):** The relationship between Users and Equipment is defined through the User Equipment junction table, which allows a User to define relationships with multiple pieces of Equipment.

Every table has its own unique identifier, which is called a primary key or PK. The foreign keys (FKs) in this dataset help to create relationships or connections to other tables; these connections define how tables are related.

The data types associated with each column in the table will follow standard database conventions. For instance, VARCHAR would be used for storing strings, INT would be for whole numbers, FLOAT would be for representing decimal numbers, ENUM will represent constrained values, DATETIME is used for representing date/time values, BOOLEAN represents 2 different states (true/false), and JSON is used to store arrays.

4.7 Sequence Diagram

A Sequence Diagram is an example of an Interaction Diagram and is used to visualize how the objects in a system will interact over time by showing the flow of messages that are exchanged in order to perform a goal or use case.

The following are the three Sequence Diagrams created for the key activities to be supported by the Smart Workout Planner & Tracker System: User Authentication, Generate a Workout Plan, and Complete a Workout Session.

4.7.1 User Registration & Login

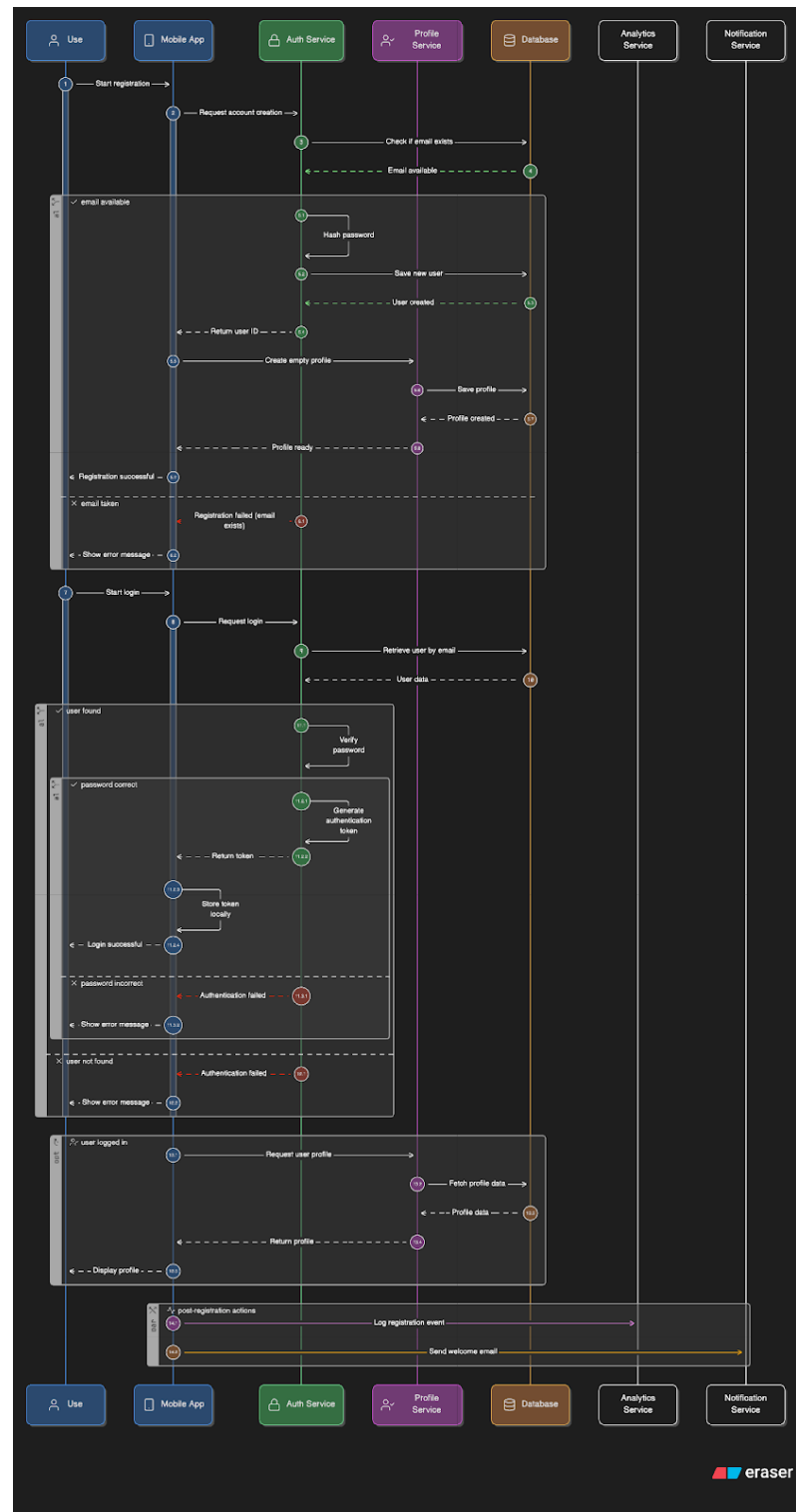


Figure 4.7: Sequence Diagram - User Registration & Login
Registration Flow:

Step	From	Interaction	Action
1-2	User	-> App -> AuthService	User submits email and password
3-4	AuthService	<-> Database	Verify email is not already registered
5	AuthService	-> AuthService	Hash password using secure algorithm
6-7	AuthService	-> Database	Save new User record
8-11	App	-> ProfileService -> Database	Create empty Profile for new user
12	App	-> User	Display registration success

Table 4.8: Flow of Parties to Complete the Registration Flow

Login Flow:

Step	From	Interaction	Action
1-2	User	-> App -> AuthService	User submits email and password
3-4	AuthService	<-> Database	Verify email is not already registered
5	AuthService	-> AuthService	Hash password using secure algorithm
6-7	AuthService	-> AuthService	Save new User record
8-9	App	-> ProfileService -> Database	Create empty Profile for new user

Table 4.9: Flow of Parties to Complete the Login Flow

The diagram uses an alternative fragment to deal with validation results (successful vs. failed). If validated, the user will be issued a token and allowed access; if not validated, an error message will be presented, prompting them to enter a new password

4.7.2 Generate Workout Plan

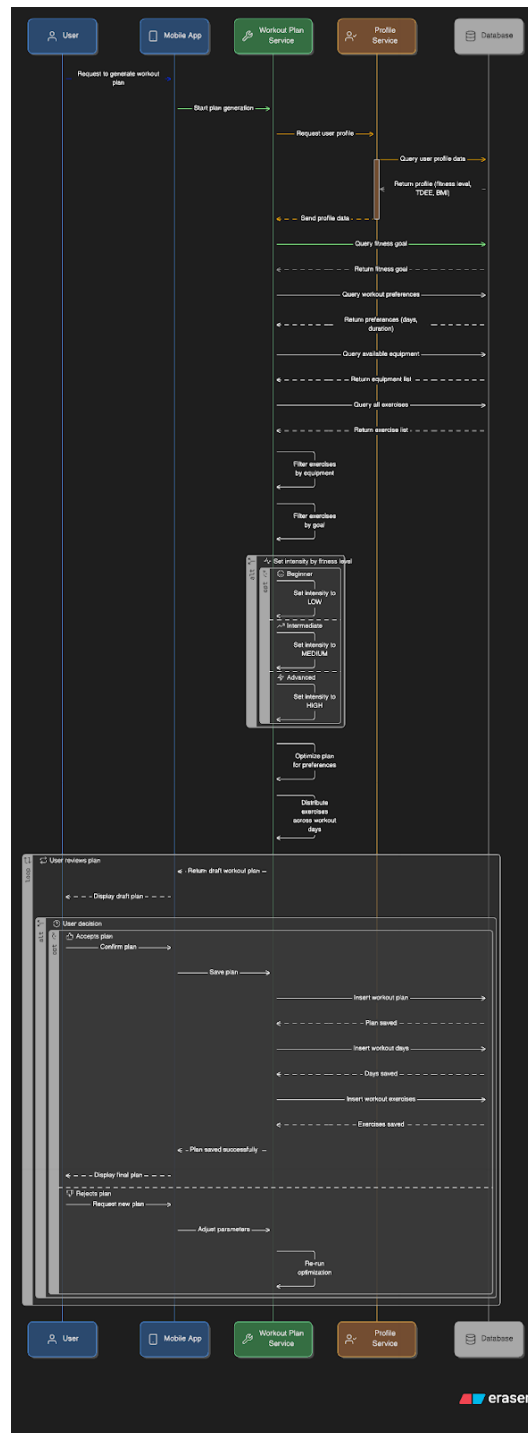


Figure 4.8: Sequence Diagram - Generate Workout Plan

The workout plan creation process is displayed in Figure 4.8, which contains the main functionality of the system. There are Five entities involved with the creation of a workout: User, Mobile Application, WorkoutPlanService, ProfileService, and Database.

Phase 1: Data Retrieval

Step	Action	Data Retrieved
1-5	Query Profile via ProfileService	fitnessLevel, tdee, bmi
6-7	Query FitnessGoal	goalType, targetDate
8-9	Query WorkoutPreference	daysPerWeek, duration, preferredTime
10-11	Query UserEquipment	Available equipment list
12-13	Query Exercise library	All exercises in database

Table 4.9.0: Flow of How Data is Retrieved

Phase 2: Filtering & Configuration

Step	Action	Data Retrieved
14	filterByEquipment()	Remove exercises requiring unavailable equipment
15	filterByGoal()	Select exercises aligned with user's fitness goal
16-18	setIntensity()	Configure volume parameters based on fitness level (Beginner → Low, Intermediate → Medium, Advanced → High)

Table 4.9.1: Flow of How Filtering & Configuration is Done

Phase 3: Optimization

Step	Action	Data Retrieved
19	applyOptimization()	Distribute exercises across workout days using algorithmic optimization

Table 4.9.2: Flow of Optimizing a Workout Plan

Phase 3: Validation & Persistence

Step	Action	Data Retrieved
20-22	Display draft plan	Present generated plan for user approval
23-30	Save plan (if accepted)	Persist WorkoutPlan, WorkoutDays, and WorkoutExercises to database
Alt	Regenerate (if rejected)	Adjust parameters and re-run optimization

Table 4.9.3: Flow of Validating a Workout Plan

The diagram outlines an alternative fragment to handle user acceptance or rejection of the generated plan. If the user does not accept the plan generated, the plan generation process may continue until the user is satisfied with the plan.

4.7.3 Complete Workout Session

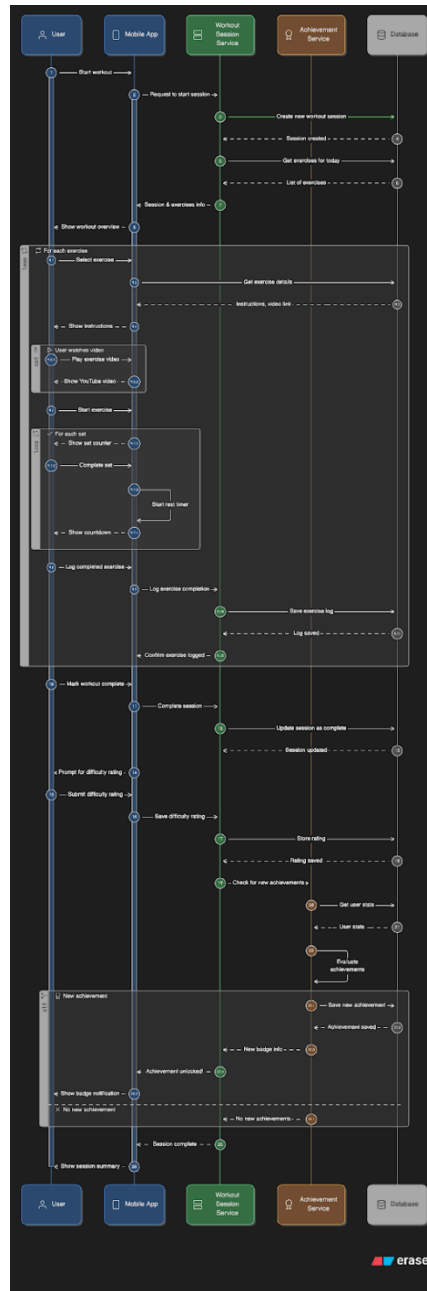


Figure 4.9: Sequence Diagram - Complete Workout Session

The workout session completion process is displayed in Figure 4.9, which illustrates the primary user activity during the workout. Five participants are involved in completing a workout session: User, Mobile Application, WorkoutSessionService, AchievementService, and Database.

The workflow proceeds through five distinct phases:

Phase 1: Session Initialization

Step	Action	Result
1-4	Start workout session	Create WorkoutSession record with completed=false
5-6	Fetch exercises	Retrieve WorkoutExercises for the selected day
7-8	Display overview	Show user the list of exercises for the session

*Table 4.9.4: Flow of Starting a Session***Phase 2: Exercise Execution Loop**

Step	Action	Result
1	View exercise	Display exercise instructions and video link
2	Watch video (optional)	Open YouTube tutorial if needed
3-4	Complete sets	Loop through each set with rest timer countdown
5-7	Log exercise	Save ExerciseLog with sets and reps completed

Table 4.9.5: Flow of Executing an Exercise

The loop fragment repeats for each exercise in the workout, and a nested loop handles individual sets within each exercise.

Phase 3: Session Completion

Step	Action	Result
1-3	Mark complete	Update WorkoutSession record (completed = true)

*Table 4.9.6: Flow of Completing a Session***Phase 4: Difficulty Rating**

Step	Action	Result
1-2	Prompt rating	Display 1-5 difficulty scale
3-5	Save rating	Store difficultyRating in WorkoutSession

Table 4.9.7: Flow of Rating a Workout's Difficulty

Phase 5: Achievement Check

Step	Action	Result
1-3	Query user stats	Retrieve workout streaks, totals, and milestones
4	Evaluate criteria	Check if any achievement conditions are met
5-8	Award badge (if earned)	Save UserAchievement and display notification

Table 4.9.8: Flow of Achievement Checking

Video Viewing and Achievement Award interaction types use Optional Fragments in the Sequence Diagram, respectively, to represent the conditionality of these interactions.

The Three Sequence Diagrams Represent the Full User Journey of the Smart Workout Planner And Tracker From Initial Registration, Authentication, Generation of Personalized Workout Plan Using Algorithmic Optimization During the Guided Workout Execution, and Progress Tracking and Gamification. These Diagrams Illustrate How Each Layer (Presentation; Business Logic; Data Persistence) of the System Will Interact With Each Other for Implementing the System.

4.8 Interface Design

Interface Design as presented provides the visual layout and user interaction patterns for the Smart Workout Planner & Tracker Application. The design emphasizes simplicity, clarity, and being user friendly and suitable for a fitness application targeting users between 18-40 Years old.

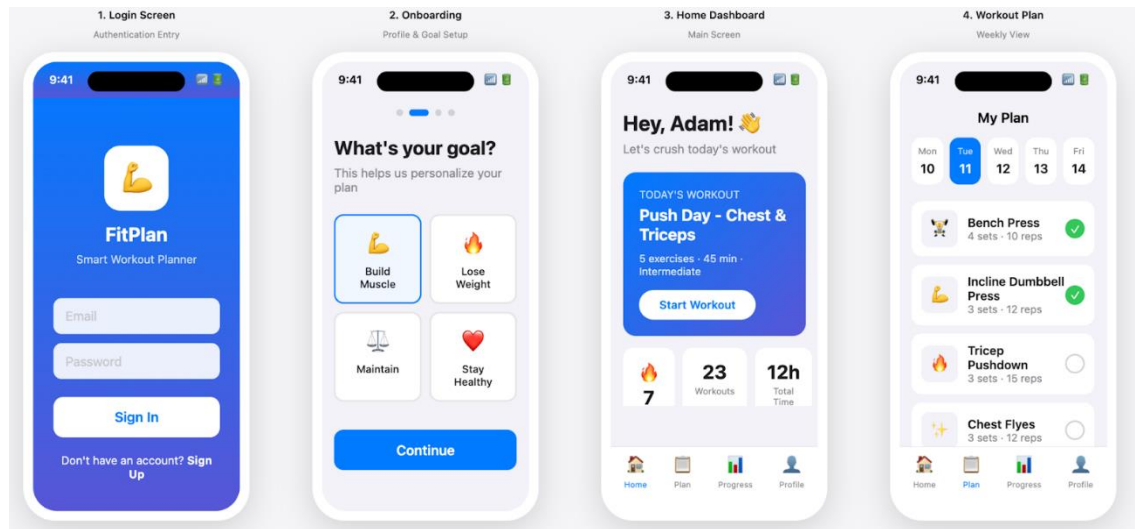


Figure 4.11: Interface Design Mockups for Login, Setting Profile and Goals, Home Dashboard, View Workout Plan

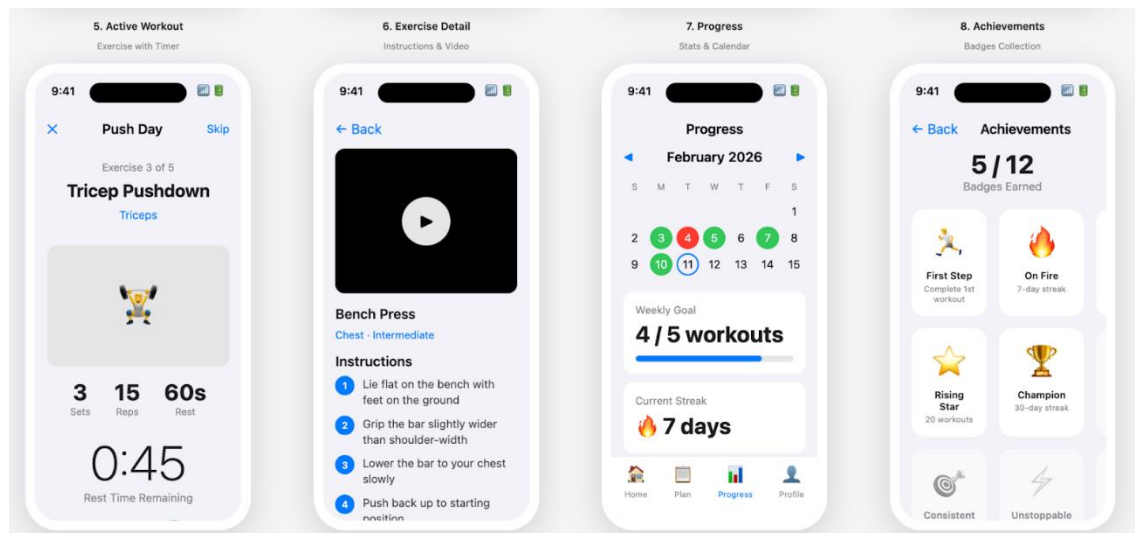


Figure 4.12: Interface Design Mockups for Active Workouts, Exercise Details, Tracking Progress, and View Achievements

Figure 4.11 and Figure 4.12 presents eight key screens representing the core user journey through the application:

Screen	Purpose	Key Features
--------	---------	--------------

Login	User authentication	Email/password input, sign up link, branded header
Onboarding	Profile and goal setup	Progress indicators, selectable goal options (Build Muscle, Lose Weight, Maintain, Stay Healthy)
Home Dashboard	Main entry point	Personalized greeting, today's workout card with quick start, streak and workout statistics
Workout Plan	Weekly schedule view	Day selector, exercise list with sets/ reps, completion checkmarks
Active Workout	Exercise execution	Current exercise display, sets/ reps information, rest timer with countdown, completion controls
Exercise Detail	Instructions and guidance	Video placeholder with play button, step-by-step instructions, muscle group tags
Progress	Statistics and history	Calendar view with workout indicators (completed/missed), weekly goal progress bar, streak counter
Achievements	Gamification badges	Badge grid showing earned and locked achievements with descriptions

Table 4.9.9: Description of Each Screens From Mockup Interface

The design of the interface follows these basic principles

- **Consistent Navigation:** A tab bar located at the bottom of the screen provides quick access to Home, Plan, Progress and Profile on each of the main screens.
- **Visual Hierarchy:** Blue gradient cards highlight primary actions (e.g., "start workout") while the secondary information is displayed in muted grey tones.
- **Progress Feedback:** Check marks, progress bars and streak counters help to show immediate evidence of user achievements.
- **Minimal Input:** Choosing from options (e.g.goals, days) reduces the amount of typing required when on-boarded or navigating through the app.
- **Motivational Elements:** By encouraging users through use of emojis, achievement badges and inspirational phrases (e.g. "Let's crush today's workout!"), users experience greater motivation to use the app.

The above listed designs are intended to serve as the foundation for the Front-End Development of the application, while providing cohesion and intuition for the user experience of the fitness journey from on-boarding to completion of fitness activities.

4.9 Summary

Through the seven diagrams or designs collectively (separate diagrams will be discussed in the following sections) presented in this chapter, a full system design has been developed for the Smart Workout Planner & Tracker application.

The use case diagram outlines the 18 expected use cases as defined by six areas of functionality and depicts the application's functional scope from the user's perspective.

The activity diagrams provide a description of the user process flows within these expected use cases that model the workflow for the user onboarding process and the generation of a prescribed workout plan; thus, indicating areas of decision-making within these processes.

The class diagram outlines the class definition of 15 entity classes utilized in a future object-oriented implementation, giving a description of each entity class's properties, methods, and relationships to other entity classes.

The entity relationship diagram depicts the aforementioned 15 entity classes as they have been translated into a database schema with the required cardinality constraints for the database Firestore.

The sequence diagrams illustrate the dynamic interaction of each entity class for the authentication process, the plan generation process, and the process of performing a workout, demonstrating the message flow between the various system components involved in each of these processes.

Chapter 5: Implementation Plan

This chapter details how the Smart Workout Planner and Tracker application will be developed, tested, and deployed. The Implementation phase will run for 14 weeks, starting on March 30, 2026, and ending on July 5, 2026, using an iterative development approach with separate front-end and back-end work streams running simultaneously throughout.

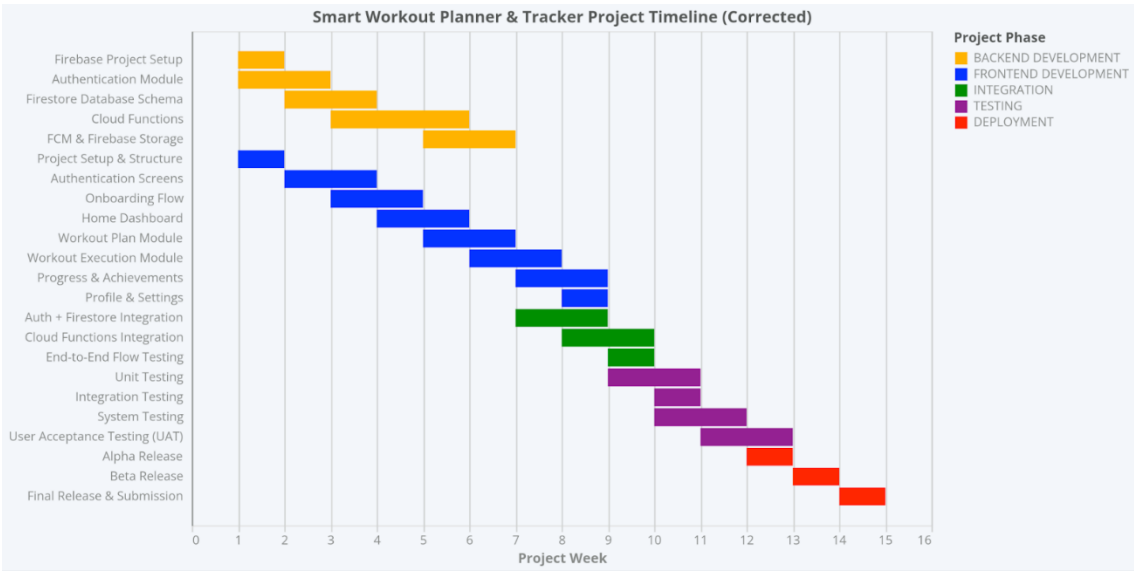


Figure 5.1: Smart Workout Planner & Tracker Project Timeline (Gantt Chart)

5.1 Development Phase (Front-end and back-end coding)

There will be three parallel development tracks, one for developing the backend (Firebase services), one for developing the front-end (Flutter application), and one for integrating both.

1. Backend Development

Task	Description	Deliverables
------	-------------	--------------

Firestore Project Setup	Initialize Firestore project, configure environment	Firestore project, security rules
Authentication Module	Implement Firestore Auth (register, login, logout, password reset)	AuthService functions
Firestore Database	Create collections, define document schemas, set up indexes	Database schema, security rules
Cloud Functions - Plan Generation	Implement workout generation algorithm	generateWorkoutPlan() function
Cloud Functions - Achievements	Implement badge checking logic	checkAchievements() function
Cloud Functions - Reminders	Implement scheduled notifications	sendReminder() function
Firestore Storage	Configure storage for profile pictures and badge icons	Storage bucket, upload rules
FCM Setup	Configure push notifications	Notification service

Table 5.1: Backend Development Process

2. Frontend Development

Module/Task	Screens/Views	Count
Project Setup	Initialize Flutter project, configure dependencies, folder structure	N/A

Authentication Screens	Login, Register, Forgot Password	3 screens
Onboarding Flow	Personal info, Fitness goal, Equipment, Preferences	4 screens
Home Module	Dashboard with today's workout and quick stats	1 screen
Workout Plan Module	Weekly plan view, Daily workout view	2 screens
Workout Execution Module	Active workout, Rest timer, Exercise detail, Rate difficulty	4 screens
Progress Module	Calendar history, Statistics	2 screens
Achievements Module	Badges display	1 screen
Profile & Settings	Profile view/edit, Reminder settings, App settings	3 screens

Table 5.2: Frontend Development Process

3. Integration

Task	Description	Deliverables
Auth Integration	Connect login/register screens to Firebase Auth	Connected UI and Auth Service

Firestore Integration	Connect all screens to Firestore collections	Data Persistence Layer Connected
Cloud Functions Integration	Connect plan generation and achievement triggers	Triggers for Remote Logic Implemented
FCM Integration	Connect reminder notifications to app	Push Notifications Working
End-to-End Flow Testing	Verify complete user journey from registration to workout completion	Verified Complete User Flow

Table 5.3: Integration Process

5.2 Testing Phase

The testing stage of the project consists of several progressive stages, from component testing to user acceptance testing.

5.2.1 Unit Testing

Unit Testing ,is the process of examining whether a single method works as expected and that it executes properly when called upon from another method. The unit tests written for this program will be written using the flutter-test framework provided with the Flutter SDK, and using the Local Emulator Suite provided with Firebase.

Test ID	Function	Test Description	Expected Result
UT-01	calculateBMI()	Input: weight=70kg, height=175cm	Returns 22.86
UT-02	calculateTDEE()	Input: male, 25yrs, 70kg, 175cm, moderate activity	Returns ~2,500 kcal

UT-03	hashPassword()	Input: "password123"	Returns hashed string (non-reversible)
UT-04	verifyPassword()	Input: correct password vs hash	Returns true
UT-05	verifyPassword()	Input: incorrect password vs hash	Returns false
UT-06	filterByEquipment()	Input: exercises[], equipment=[DUMBBELL]	Returns only dumbbell exercises
UT-07	setIntensity()	Input: fitnessLevel=BEGINNER	Returns low volume parameters
UT-08	checkAchievement()	Input: user with 7-day streak	Returns "On Fire" badge

Table 5.4: Unit Testing List

5.2.2 Integration Testing

Integration Testing, is the process of verifying whether groups of methods or classes communicate properly. In this program, integration testing will be performed by ensuring a correct flow of data between the frontend screens, Firebase services, and Cloud Functions.

Test ID	Integration Flow	Test Description	Expected Result
IT-01	Auth → Firestore	Register New User	User document created in Firestore
IT-02	Auth → Profile	Complete On Boarding	Profile, FitnessGoal, WorkoutPreference documents created

IT-03	Profile → Cloud Functions	Register Workout Plan	generateWorkoutPlan() triggered, WorkoutPlan created
IT-04	WorkoutPlan → WorkoutDay	View Plan	WorkoutDays subcollection retrieved correctly
IT-05	WorkoutSession → ExerciseLog	Complete Workout	ExerciseLog documents saved, session marked complete
IT-06	WorkoutSession → Achievements	Complete Workout	checkAchievements() triggered, badge awarded if criteria met
IT-07	Reminder → FCM	Scheduled Reminder Time	Push notification delivered to device

Table 5.5: Integration Testing List

5.2.3 System Testing

System Testing, is the process of testing the final product against both functional and non-functional requirements using black-box techniques.

Functional Testing Checklist:

Test ID	Feature	Test Case	Pass/Fail
ST-01	Register	User can register with valid email and password	☐
ST-02	Login	User can login with correct credentials	☐
ST-03	Login	User sees error with incorrect credentials	☐

ST-04	Onboarding	User can complete all onboarding steps	☐
ST-05	Plan Generation	System generates workout plan based on user inputs	☐
ST-06	View Plan	User can view weekly workout plan	☐
ST-07	Swap Exercise	User can swap exercise in plan	☐
ST-08	Regenerate Plan	User can regenerate workout plan	☐
ST-09	View Exercise	User can view exercise instructions	☐
ST-10	Watch Video	User can open YouTube video link	☐
ST-11	Rest Timer	Timer counts down correctly	☐
ST-12	Complete Workout	User can mark workout complete	☐
ST-13	Rate Difficulty	User can rate workout difficulty (1-5)	☐
ST-14	View History	User can view workout history in calendar	☐
ST-15	View Progress	User can view progress statistics	☐
ST-16	View Badges	User can view earned achievement badges	☐
ST-17	Receive Reminder	User receives push notification reminder	☐
ST-18	Logout	User can logout successfully	☐

Table 5.6: Functional Testing Checklist

Non-Functional Testing:

Test ID	Requirement	Test Case	Expected Result
NFT-01	Performance	App launch time	< 3 seconds
NFT-02	Performance	Screen transition	< 1 second
NFT-03	Performance	Plan generation	< 5 seconds
NFT-04	Offline	Access cached workout plan	Plan viewable without internet
NFT-05	Usability	Complete core user journey	No critical errors

Table 5.7: Non-Functional Testing List

5.2.4 User Acceptance Testing (UAT)

User acceptance testing (UAT) provides a means of testing an application with actual users to confirm that it meets their anticipated needs and requirements. UAT will take place with ten (10) people from the intended target audience (i.e., university students & young adults aged 18-40).

UAT Procedure:

Step	Activity	Duration
1	Brief participants on app purpose and testing objectives	5 minutes
2	Participants install app via Firebase App Distribution	5 minutes
3	Participants complete assigned tasks independently	30 minutes

4	Participants complete System Usability Scale (SUS) questionnaire	5 minutes
5	Optional: Verbal feedback and observations	10 minutes

Table: 5.8: UAT Procedure List

System Usability Scale (SUS) Questionnaire:

#	Statement	Strongly Disagree (1)	2	3	4	Strongly Agree (5)
1	I think that I would like to use this app frequently	☐	☐	☐	☐	☐
2	I found the app unnecessarily complex	☐	☐	☐	☐	☐
3	I thought the app was easy to use	☐	☐	☐	☐	☐
4	I think I would need technical support to use this app	☐	☐	☐	☐	☐
5	I found the various functions were well integrated	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this app	☐	☐	☐	☐	☐
7	I imagine most people would learn to use this app quickly	☐	☐	☐	☐	☐
8	I found the app very cumbersome to use	☐	☐	☐	☐	☐
9	I felt very confident using the app	☐	☐	☐	☐	☐

10	I needed to learn a lot before I could get going with this app	☐	☐	☐	☐	☐
----	--	--------------------------	--------------------------	--------------------------	--------------------------	--------------------------

Table: 5.9: SUS Questionnaire List

SUS Score Interpretation:

Score Range	Rating
0 - 50	Not Acceptable
51 - 70	Marginal
71 - 85	Good
86 - 100	Excellent

Table: 5.9.1: SUS Score Interpretation

5.3 Deployment Phase

The Deployment Phase will be performed using a staged rollout framework which will help to avoid risk and allow for feedback to be gathered on an iterative basis.

5.3.1 Deployment Stages

Stage	Timeline	Audience	Purpose
Alpha Release	Week 12 (15-21 Jun 2026)	Developer only	Internal testing, bug identification

Beta Release	Week 13 (22-28 Jun 2026)	10 UAT participants	User acceptance testing, feedback collection
Final Release	Week 14 (29 Jun - 5 Jul 2026)	Evaluators / Presentation	Project submission and demonstration

Table: 5.9.2: Deployment Stages

5.3.2 Deployment Platform

Component	Platform	Method
Flutter App (Android)	Firebase App Distribution	APK distributed via email/link
Flutter App (iOS)	Firebase App Distribution	TestFlight (if iOS build required)
Backend	Firebase (Cloud)	Automatic (serverless)

Table: 5.9.3: Deployment Platforms

5.3.3 Alpha Release

Activity	Description
Build Generation	Generate release APK with production Firebase configuration
Internal Testing	Developer performs full regression testing
Bug Fixing	Address critical and high-priority bugs

Performance Check	Verify app performance on target devices
--------------------------	--

Table: 5.9.4: Alpha Release Activity List

Alpha release is scheduled for Week 12 to allow sufficient buffer (2 weeks) before final submission for bug fixes and refinements.

5.3.4 Beta Release

Activity	Description
Distribution	Upload APK to Firebase App Distribution
UAT Execution	Conduct user acceptance testing with 10 participants
Feedback Collection	Gather SUS scores and verbal feedback
Bug Triage	Prioritize and fix issues reported by testers

Table: 5.9.5: Beta Release Activity List

Beta release is scheduled one week before final submission to allow time for incorporating user feedback while maintaining a stable build.

5.3.5 Final Release

Activity	Description
Final Build	Generate final release APK incorporating beta fixes
Documentation	Complete user manual and technical documentation

Submission	Submit project deliverables
Demonstration	Present application to evaluators

Table: 5.9.6: Final Release Activity List

5.3.6 Deployment Checklist

#	Item	Status
1	Firebase production environment configured	☐
2	Security rules tested and deployed	☐
3	Cloud Functions deployed to production	☐
4	Release APK signed and generated	☐
5	App Distribution configured	☐
6	Test users invited	☐
7	Crash reporting enabled (Firebase Crashlytics)	☐
8	Analytics enabled (Firebase Analytics)	☐

Table: 5.9.7: Deployment Checklist

5.5 Summary

This implementation plan offers a structured framework for developing, testing, and delivering the Smart Workout Planner & Tracker application within fourteen (14) weeks using parallel methods to utilize resources efficiently, progressive testing methods for

quality assurance at each development step, and a staged deployment model to minimize risk and allow for continuation of iterative development through user feedback with the ultimate goal of providing a stable and reliable final release version to be used for project submission

Chapter 6: Conclusion

6.1 Summary

The purpose of this project is to create a Smart Workout Planner & Tracker mobile application that solves the problem of exercise adherence by offering personalized, evidence-based solutions. The following tasks were completed in the FYP1 phase:

Literature Review (Chapter 2): A thorough review of twenty academic articles results in establishing an empirical and theoretical basis for the application. The literature highlights the serious drop out rates from exercise (45-80% in four years), the importance of providing users with personalised options to choose from rather than having to use pre-set templates, and that gamification and smart reminders are effective for increasing user engagement.

Requirements Analysis (Chapter 3): Through an online survey conducted with one hundred one respondents, confirmation was gained about the intended audience (90.3% aged between 18-24; 79.0% university students) and identified their needs. The following barriers were confirmed as the top three reasons stopping them from engaging with physical activity; (Fatigue 58.1%, lack of time 55.6%, lack of motivation 55.6%). The highest rated feature (87.1%) was that users want to track and visualise their progress.

System Design (Chapter 4): The design of the application is based upon a three layer model (Flutter front end; Firebase back end; YouTube API integration). The key deliverables include; Use Case Diagrams (18 Use Cases), Class Diagrams (15 Entities), Sequence Diagrams, and Interface Mockups (8 Screens).

Implementation Plan (Chapter 5): A Developer's Plan outlining a 14 week completion window was developed and runs from March – July 2026 with specified milestones for back end, front end and integration testing/deployment.

6.2 Expected Contributions

Upon completion, this project is expected to contribute:

1. An operational mobile application that creates a custom workout plan, depending on user's limitations as to: their fitness level, the equipment at their disposal, the time they have available, and their fitness goals.
2. The provision of evidence-based techniques for the incorporation of motivation elements such as: progress tracking, badges for achievement, and notification reminders with optimal times.
3. A no-cost, accessible alternative that addresses the cost barrier found in the literature

6.3 FYP2 Plan

The objectives for this next step (FYP2) of the project are to focus on the following areas:

1. **Implementation:** A fully developed Flutter application and Firebase backend as per the defined construction.
2. **Testing:** Execution of unit testing, integration testing, and user acceptance testing with 10-15 users from the target population.
3. **Evaluation:** Performing a usability audit using the System Usability Scale (SUS) and collecting qualitative feedback.
4. **Refinement:** Revising and making changes to the design based on the testing results.

References

1. Alberts, L., Lyngs, U., & Lukoff, K. (2024). Designing for sustained motivation: A review of self-determination theory in behaviour change technologies. *Interacting with Computers*, iwae040. <https://doi.org/10.1093/iwc/iwae040>
2. Brown, D. M. Y., Bray, S. R., Beatty, K. R., & Kwan, M. Y. W. (2024). Key influences on university students' physical activity: A systematic review using the COM-B model and theoretical domains framework. *International Journal of Behavioral Nutrition and Physical Activity*, 21(1), 43.
3. Du, J., & Deng, Q. (2025). Exploring exerciser experience with personalized workout plans in AI-powered fitness apps: A content analysis of online reviews. [Manuscript in preparation].
4. Eriksson, R., Nicander, J., Johansson, M., & Mattsson, C. M. (2021). Generating weekly training plans in the style of a professional swimming coach using genetic algorithms and random trees. In *Proceedings of the International Conference on Computational Intelligence in Sports*. Springer.
5. Haroon, R. (2025). *The effectiveness of Facebook and Fitbit in motivating Finnish adults to stay physically active* [Master's thesis, JAMK University of Applied Sciences].
6. Ikegaya, M., Foo, J. C., Murata, T., Oshima, K., & Kim, J. (2025). Using personalized intervention criteria in a mobile just-in-time adaptive intervention for increasing physical activity in university students: Pilot study. *JMIR Human Factors*, 12, e66750. <https://doi.org/10.2196/66750>
7. Kerr, J. (2025). *Optimization and personalization of fitness routines using mathematical modeling* [Senior project].
8. Kim, S., & Lee, Y. (2023). Text-mining analysis of Fitbit app reviews using Herzberg's two-factor theory. *Behavioral Sciences*, 13(10), 782. <https://doi.org/10.3390/bs13100782>
9. Kunkel, J., Donkers, T., Michael, L., Barbu, C. M., & Ziegler, J. (2023). The effects of push notifications promoting gameful affordances on consumer engagement with a mobile app. *Computers in Human Behavior*.
10. Lang, S., McLelland, C., MacDonald, D., & Hamilton, D. F. (2022). Do digital interventions increase adherence to home exercise rehabilitation? A systematic review of randomised controlled trials. *Archives of Physiotherapy*, 12, 24. <https://doi.org/10.1186/s40945-022-00148-z>
11. Lunde, P., Nilsson, B. B., Bergland, A., Kværner, K. J., & Bye, A. (2020). Long-term follow-up with a smartphone application improves exercise capacity post cardiac rehabilitation: A randomized controlled trial. *European Journal of Preventive Cardiology*, 27(16), 1776–1785.

12. Mahmood, A., Nayak, P., Deshmukh, A., English, C., Manikandan, N., Solomon, J. M., & Unnikrishnan, B. (2023). Measurement, determinants, barriers, and interventions for exercise adherence: A scoping review. *Journal of Bodywork & Movement Therapies*, 33, 95–105.
<https://doi.org/10.1016/j.jbmt.2022.09.014>
13. Mustafa, A. S., Ali, N., Dhillon, J. S., Alkawsi, G., & Baashar, Y. (2022). User engagement and abandonment of mHealth: A cross-sectional survey. *Healthcare*, 10(2), 221.
14. Pérez-Jorge, D., Martínez-Murciano, M. C., Gutiérrez-Barroso, J., Barrios-Fernández, S., & González-Herrera, A. I. (2024). Gamified physical exercise and its effects on adolescents' mental health: A systematic review. *Healthcare*, 12(1), 124.
<https://doi.org/10.3390/healthcare12010124>
15. Shan, P. C. J., & Suparjoh, S. (2022). Development of mobile application for health and fitness guidance. *International Journal of Interactive Mobile Technologies*.
16. Sun, J., Wang, Y., & Chen, Q. (2021). Personalized mobile fitness apps with exercise recommendations: The role of personality and exercise frequency. *JMIR mHealth and uHealth*, 9(9), e19941.
<https://doi.org/10.2196/19941>
17. Wang, C., Zhu, X., Hong, J. C., & Zheng, D. (2021). Reinforcement learning to send reminders at right moments in smartphone exercise application: A feasibility study. *International Journal of Environmental Research and Public Health*, 18(11), 6059.
<https://doi.org/10.3390/ijerph18116059>
18. Wunsch, K., Eckert, T., Fiedler, J., & Woll, A. (2020). Just-in-time adaptive interventions in mobile physical activity interventions: A synthesis of frameworks and future directions. *The European Health Psychologist*, 22(4), 834–841.
19. Yang, Y., & Koenigstorfer, J. (2020). Determinants of physical activity maintenance during the Covid-19 pandemic: A focus on fitness apps. *Translational Behavioral Medicine*, 10(4), 835–842.
<https://doi.org/10.1093/tbm/ibaa086>
20. Zhu, Y., Dailey, S. L., Kreitzberg, D., & Bernhardt, J. (2024). Digital behavior change intervention designs for habit formation: Systematic review. *Journal of Medical Internet Research*, 26, e54375.
<https://doi.org/10.2196/54375>

Appendix A



**FACULTY OF
COMPUTING
& INFORMATICS**

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 4/12/2025	Meeting No.: 1
Meeting Mode: Face-to-Face	
Project ID: 771	Project Type: Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

--	--

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: **Problem Formulation and Project Planning** / ~~Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Conducted initial project introduction with supervisor
- Presented draft use cases outlining core application functionalities
- Proposed preliminary functional requirements for the fitness tracking system
- Outlined non-functional requirements including performance and usability considerations
- Discussed general workflow approach for FYP execution, covering documentation phases and required diagrams (use case diagram, UML diagrams, ERD)
- Explored potential technology stacks, with initial preference towards Flutter with Dart, containerised using Docker

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning /~~ Background Study or Literature Review / ~~Requirement Analysis or Theoretical Framework / Design or Research Methodology /~~ ~~Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):*Immediate:*

- Develop a project timeline with tasks and action items using Microsoft Excel
- Set up a GitHub repository to manage and version-control all project artefacts

Ongoing:

- Commence literature review by identifying and gathering a minimum of 20 academic papers relevant to fitness tracking, personalisation algorithms, and mobile application development
- Revisit and refine drafted use cases and requirements to abstract complex functionalities, ensuring reduced risk during development phase

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

PROBLEMS:

1. Uncertainty in selecting the appropriate technology stack due to limited experience in mobile application development. React Native offers familiarity from prior React web development, whereas Flutter with Dart presents a valuable learning opportunity.
2. Overwhelmed by the volume of academic papers required for literature review (minimum 20 papers).

SOLUTIONS:

1. Conduct a comparative analysis of both frameworks, evaluating factors such as performance, community support, learning curve, and suitability for cross-platform fitness applications. Finalise decision by next meeting.
2. Strategise paper selection by categorising into themes: (1) fitness tracking methodologies, (2) personalisation and adaptive algorithms, (3) mobile app UI/UX for health applications, and (4) similar existing systems. Use Google Scholar and IEEE Xplore as primary sources.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)





 Supervisor's Signature



 Student's Signature

.....
 Co-Supervisor's Signature
 (if applicable)

.....
 Company Supervisor's Signature
 (if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can

have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 11/12/2025	Meeting No.: 2
Meeting Mode: Online	
Project ID: 771	Project Type: Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

--	--

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / **Background Study or Literature Review** / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Established GitHub repository to centralise and version-control all project artefacts and progress
- Developed project timeline in Microsoft Excel outlining tasks and milestones
- Created draft Google Docs workspace for report writing
- Completed Chapter 1 (Introduction) draft comprising the following sections:
 1. Project Overview
 2. Problem Statement
 3. Project Objectives
 4. Project Limitations
 5. Methodology
 6. Target Audience
 7. Summary
- Acquired 12 academic papers for literature review

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~ / **Background Study or Literature Review**

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Source remaining academic papers to meet the minimum requirement of 20 papers
- Commence writing Chapter 2 (Literature Review) of the FYP report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

PROBLEMS:

1. Previously faced difficulty in identifying relevant academic papers for the literature review due to the broad scope of the project.
2. Technology stack selection remains undecided between React Native and Flutter, both intended to be containerised with Docker.

SOLUTIONS:

1. Resolved by reframing the search strategy around the target audience's core challenges in fitness adoption, specifically: lack of motivation, time constraints, and insufficient knowledge on initiating workout routines. By aligning paper searches with these factors and the application's proposed features, the majority of required papers were successfully identified.
2. Continue evaluating both frameworks. Plan to finalise decision upon completing further research or based on supervisor's recommendation. Decision to be confirmed by next meeting.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)





 Supervisor's Signature



 Student's Signature

.....
 Co-Supervisor's Signature
 (if applicable)

.....
 Company Supervisor's Signature
 (if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



**FACULTY OF
COMPUTING
& INFORMATICS**

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 8/1/2026	Meeting No.: 3
Meeting Mode: Online / Face-to-Face	
Project ID: 771	Project Type: Research-based / Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning /~~ Background Study or Literature Review / ~~Requirement Analysis or Theoretical Framework / Design or Research Methodology /~~ ~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Successfully acquired remaining academic papers, completing the minimum requirement of 20 papers for literature review
2. Commenced systematic review of all collected papers
3. Conducted integrated literature review synthesizing findings across multiple sources
4. Held detailed discussion with supervisor regarding data requirements for Integer Linear Programming (ILP) algorithm implementation
5. Identified key data parameters needed for automated workout routine generation system
6. Explored mathematical modeling approaches for workout optimization

2. WORK TO BE DONE

Tasks: ~~Problem Formulation and Project Planning~~ / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / ~~Design or Research Methodology~~ / ~~Prototype Development or Proof of Concept~~ / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Refine Chapter 2 (Literature Review) by incorporating mathematical equations and formulas to strengthen theoretical justification
- Integrate relevant algorithms and computational models into literature review analysis
- Draft survey requirements and questionnaire design for Chapter 3 (Methodology/Requirements Analysis)
- Identify target respondent demographics and sample size for survey distribution
- Prepare survey instruments for supervisor review

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Problem

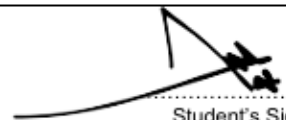
1. Uncertainty regarding supervisor's request to incorporate mathematical equations into the literature review. The purpose and placement of these equations within the integrated review structure remains unclear.
2. Need to transition from theoretical literature review to practical requirements gathering through surveys, requiring clear understanding of what data to collect.

Solution

1. Proceed with adding relevant equations (e.g., ILP optimization formulas, fitness calculation models, personalization algorithms) to literature review sections where applicable. Schedule follow-up clarification with supervisor if needed to ensure equations appropriately support the review's legitimacy and academic rigor.
2. Leverage insights from data requirement discussions with supervisor to inform survey design. Align survey questions with identified ILP data parameters to ensure collected user data will support the proposed workout generation algorithm.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)


.....
Supervisor's Signature


.....
Student's Signature

.....
Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide

the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 23/1/2026	Meeting No.: 4
Meeting Mode: Online / Face-to-Face	
Project ID: 771	Project Type: Research-based / Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Completed Chapter 2 (Literature Review) with integrated mathematical equations and formulas
2. Drafted comprehensive functional requirements outlining core system capabilities
3. Drafted non-functional requirements covering performance, usability, security, and scalability considerations
4. Drafted user requirements based on target audience analysis
5. Presented all draft requirements to supervisor for review and received approval to proceed
6. Created initial prototype design in Figma demonstrating user interface and interaction flows
7. Developed architecture diagram illustrating system components and their relationships
8. Designed preliminary class diagram mapping object-oriented structure of the application
9. Prepared foundation for Chapter 4 (System Design) with visual design artifacts

2. WORK TO BE DONE

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Conduct user requirements survey to validate and refine drafted requirements
2. Distribute survey and collect minimum of 20 responses from target audience
3. Analyze survey results to confirm or adjust functional, non-functional, and user requirements
4. Finalize requirements specifications based on survey findings
5. Refine and complete class diagrams incorporating validated requirements
6. Finalize Figma prototype based on user feedback and confirmed requirements
7. Complete Chapter 3 (Methodology/Requirements Analysis) including survey methodology, results, and analysis
8. Complete Chapter 4 (System Design) with finalized diagrams and design specifications

3. PROBLEMS ENCOUNTERED AND SOLUTIONS	
<u>Problem</u>	
1. Previously encountered confusion regarding the integration of mathematical equations into literature review. 2. All requirements and design artifacts are currently in draft form and cannot be finalized until user survey validation is completed, creating dependency on survey response collection.	
<u>Solution</u>	
1. Successfully resolved by incorporating relevant optimization formulas, ILP modeling equations, and algorithmic representations into the literature review, strengthening its academic foundation. Supervisor confirmed adequacy of the approach. 2. Prioritize survey development and distribution immediately to meet the minimum 20-response requirement. Design survey questions to specifically validate drafted requirements (functional, non-functional, user). Allocate sufficient time for response collection to avoid delays in finalizing Chapter 3 and Chapter 4 documentation.	

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)


.....
Supervisor's Signature


.....
Student's Signature

.....
Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



**FACULTY OF
COMPUTING
& INFORMATICS**

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 9/2/2026	Meeting No.: 5
Meeting Mode: Online / Face-to-Face	
Project ID: 771	Project Type: Research-based / Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Completed Chapter 3 (Methodology/Requirements Analysis) with comprehensive survey analysis
2. Successfully conducted user requirements survey, collecting 124 responses (significantly exceeding minimum requirement of 20 responses)
3. Finalized and validated functional requirements based on survey data analysis
4. Finalized and validated non-functional requirements incorporating user feedback
5. Finalized and validated user requirements aligned with target audience needs
6. Completed all required system design diagrams:
 - Use case diagram illustrating system actors and interactions
 - Class diagram depicting object-oriented structure and relationships
 - Architecture diagram showing system components and layers
 - Activity diagram mapping workflow processes
 - Entity Relationship Diagram (ERD) defining database schema
 - Sequence diagram demonstrating interaction flow between system components
7. Integrated initial UI designs and interface mockups into Chapter 4
8. Commenced writing Chapter 5 (Implementation/System Development)

2. WORK TO BE DONE

~~Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept /~~ **Draft Report Completion**

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Complete Chapter 5 (Implementation/System Development) detailing development approach and technical implementation
2. Draft Abstract summarizing entire project scope, methodology, and expected outcomes
3. Write Acknowledgements section recognizing supervisor guidance and contributors
4. Finalize scope of system functionalities to be implemented
5. Prepare for transition to FYP2 development phase

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Problem

1. Faced difficulty in finalizing system scope while drawing comprehensive diagrams. Tension exists between implementing all planned features versus reducing complexity by abstracting or removing certain advanced functionalities.
2. Concern about over-engineering the system with complex functions that may pose implementation risks or exceed realistic development timeline constraints.

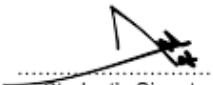
Solution

1. Conduct critical evaluation of each proposed function against project objectives and available development time. Prioritize core features that directly address primary problem statement and user needs identified in survey. Consider implementing a Minimum Viable Product (MVP) approach with essential features, relegating complex functions to future enhancements or optional stretch goals.
2. Consult with supervisor to determine which complex functions can be simplified, abstracted, or deferred without compromising project integrity. Update diagrams and requirements documentation to reflect refined scope, ensuring alignment between design artifacts and feasible implementation plan. This strategic reduction will improve project manageability while maintaining academic and practical value.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



 Supervisor's Signature



 Student's Signature

.....
 Co-Supervisor's Signature
 (if applicable)

.....
 Company Supervisor's Signature
 (if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.

2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



**FACULTY OF
COMPUTING
& INFORMATICS**

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 10/2/2026	Meeting No.: 6
Meeting Mode: Online / Face-to-Face	
Project ID: 771	Project Type: Research-based / Application-based
Project Title : Personalized Fitness Tracker & Smart Workout Planner Mobile Application	
Student ID : 1211111950	Student Name: ADAM FIKRI BIN MOHD LOTFI
Student Programme and Specialisation: B.C.S (Hons.) Software Engineering	
Supervisor Name: Nur Haifa binti Mohd Fathil	Co-Supervisor Name: (if applicable) N/A
Collaborating Company: (if applicable) N/A	Company Supervisor Name: (if applicable) N/A

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept /~~ **Draft Report Completion**

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Completed Chapter 5 (Implementation/System Development) documenting development methodology and technical approach
2. Drafted and finalized Abstract providing comprehensive project summary
3. Completed Acknowledgements section
4. Finalized all FYP1 report chapters (Chapters 1-5)
5. Performed internal review and proofreading of complete report document
6. Submitted complete FYP1 report to supervisor for Turnitin similarity check
7. Submitted report for AI detection verification to ensure academic integrity
8. Held comprehensive discussion with supervisor regarding implementation strategies for FYP2
9. Outlined development roadmap and timeline for system implementation in upcoming semester
10. Discussed technical stack finalization and development environment setup for FYP2

2. WORK TO BE DONE

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept~~ **Draft Report Completion**

(Please strike out the tasks, which are not applicable)

Details (in point form):

1. Review Turnitin similarity report and make necessary revisions if required
2. Address any AI detection concerns if flagged
3. Make final corrections based on supervisor feedback
4. Prepare final submission of FYP1 report

Planning for FYP2:

1. Finalize technology stack selection (React Native vs Flutter) based on implementation requirements
2. Set up development environment and project structure
3. Begin preliminary implementation of core system modules
4. Establish development workflow and version control practices for FYP2 phase

3. PROBLEMS ENCOUNTERED AND SOLUTIONS	
<u>Problem</u>	1. Completed FYP1 documentation phase without major outstanding issues. Previous concern regarding scope management has been addressed through diagram refinement and function prioritization. 2. Transition phase between FYP1 (documentation) and FYP2 (implementation) requires clear planning to ensure smooth continuation of project momentum.
<u>Solution</u>	1. Successfully navigated FYP1 requirements by maintaining consistent meeting schedule, proactive communication with supervisor, and systematic completion of each chapter. Turnitin submission ensures compliance with academic integrity standards. 2. Utilize break period before FYP2 semester to prepare development environment, review finalized requirements and design documents, and create detailed implementation plan. Schedule early FYP2 meeting with supervisor to establish clear milestones and deliverables for implementation phase. This preparation will enable immediate productive work when FYP2 officially begins.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Supervisor's Signature

Student's Signature

Co-Supervisor's Signature
(if applicable)Company Supervisor's Signature
(if applicable)**IMPORTANT NOTES TO STUDENTS:**

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can

have fortnightly meetings with the supervisor.

3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Appendix B



Page 2 of 109 - Integrity Overview

Submission ID: trnoid::3618:128021097

14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

- 124 Not Cited or Quoted 14%**
Matches with neither in-text citation nor quotation marks
- 16 Missing Quotations 1%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 5% Internet sources
- 4% Publications
- 13% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.